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1. Title of the Invention

Wearable Apparatus and Sensor-Responsive Method for Real-Time Obstacle Threat Prioritisation and Adaptive Multimodal Navigation Guidance Using Inertially Adaptive Detection and Staged Announcement Scheduling

2. Field / Area of Invention

The present invention relates to the field of wearable assistive apparatus and, more particularly, to a chest-worn cyber-physical apparatus that observes an indoor environment through an imaging sensor and an inertial sensor, converts each detected object into a single ordinal threat control value, and drives voice, vibrotactile and spatial-audio guidance actuators in dependence upon that value. The invention is an actuation control apparatus: its output is not a caption, a description or a screen reading, but a timed activation of physical transducers worn against the body of the wearer.

The operating environment is the interior of a dwelling, an office or an institutional building, in which the obstacle configuration changes continuously as furniture is moved, doors stand part-open and people cross the path of travel. The physical consequences of failure are direct. A missed obstacle produces a collision. An obstacle announced too late produces a collision the wearer could not avoid; and an obstacle announced too often produces a stream of actuator activity through which the wearer cannot hear the one warning that matters. Because the apparatus is worn on the chest, its imaging sensor moves with the trunk at walking pace, so the captured image is degraded by that motion at exactly the moments when detection is most needed.

The invention resolves this by inserting a sensor-responsive threat prioritisation control process and a four-stage temporal output-control state machine into the path between the detector and the actuators, and by applying a single inertial measurement at two separate points in that path. The

apparatus is constituted from commercially available elements, and the technical contribution resides in their arrangement and in the constraints under which sensed data determines actuator state, not in any individual element.

2.1 Technical Problem Addressed by the Invention

- **Motion-induced loss of detections in a body-worn optical sensor.** An imaging sensor on the trunk of a walking person undergoes translational and rotational motion at the gait frequency. The frame blurs, detector confidence for a valid object falls below a fixed acceptance threshold, and the detection is discarded at precisely the moment the wearer is moving toward the object. Permanently raising detector sensitivity is no solution, because the same setting admits noise when the wearer is stationary.
- **Absence of a quantitative basis for allocating one actuator set among many detections.** An indoor frame commonly contains five to fifteen recognised objects. The apparatus possesses one voice channel, one stereo tone channel and three vibration positions, and can therefore signal one object per event. Reviewed installations resolve this by nearest-object heuristics or detection order, neither of which accounts for whether the object lies on the path of travel or whether the wearer is moving toward it.
- **Uncontrolled actuation rate from a per-frame detection stream.** A detector on a continuous video stream produces a detection for the same object on every frame. Driving the actuators from that stream causes near-continuous vibration, contention for the single audio channel, and queueing of stale spoken messages behind current ones, so the wearer hears a warning about an object already passed.
- **Suppression schemes that cannot respond to a worsening situation.** A suppression interval applied uniformly to all repeat warnings prevents redundant output, but equally prevents re-signalling an object whose threat is rising rapidly within that interval. Such a scheme is safe against annoyance and unsafe against approach.
- **Correctly classified objects that are navigationally irrelevant.** A small object at close range occupies enough of the frame to satisfy any area-based proximity criterion while presenting no collision risk, so a cup correctly identified as a cup generates an obstacle warning under a purely geometric rule. Within the evaluated dataset the six classes most affected produced navigationally irrelevant walking-mode warnings at rates from 9.4% to 23.1%, with a mean of 15.6%.
- **Occlusion of the ear canal by the output channel.** Auditory guidance through earbuds removes the wearer's access to ambient sound, which for a visually impaired person is a primary sensing channel. An apparatus that improves obstacle awareness by degrading acoustic awareness has not improved safety.

The invention addresses this set by applying one inertial measurement both to the operating point of the detector and to the threat computation, by reducing every detection to one ordinal value that ranks it for actuation, by admitting the selected object through a four-stage state machine in which an escalation criterion overrides the per-state interval but never the global interval, by gating actuation on the class label independently of the computed value, and by delivering both audio channels through an open-ear transducer pair.

2.2 Definitions

Threat Control Value: A scalar quantity, T(o), computed for each surviving detection on each cycle as a weighted combination of four scene parameters per Equation 3. Its function is ordinal: detections are ranked by T(o) and the highest-ranked object alone may drive the actuator set. T(o) is not asserted to be a collision probability, and no protection is sought for the combination as a mathematical method in the abstract. It is defined by the sensor inputs from which it is derived and the actuator behaviour it determines.

Proximity Rank: A class-calibrated ordinal distance proxy, P(o), taking one of four values corresponding to very close, close, medium and far. Observed bounding-box area is compared against thresholds specific to the assigned class, so a bottle at 5% of frame area and a person at 35% both resolve to very close. It is distinct from apparent size S(o), the uncalibrated normalised area, which is independent of class identity.

Forward-Axis Centrality: A quantity, C(o), computed per Equation 4 from the normalised horizontal coordinate of the bounding-box centre, expressing the position of the object relative to a nominal forward travel axis represented by the horizontal centre of the captured image. It is expressly recorded that this axis is a nominal image-frame axis and not an independently reconstructed walkable path. The apparatus performs no floor segmentation.

Wearer-Motion Convergence Factor: A quantity, M(o), obtained by resolving the instantaneous inertially measured motion of the apparatus onto the angular bearing of a given detection and quantising the result to approaching, stationary-near or receding. Because the bearing term differs between objects, M(o) differs between objects in the same frame. An inertial sensor observes only the motion of the apparatus, so M(o) is a wearer-motion-derived convergence indicator for static obstacles and not a measurement of relative object velocity; for

an independently moving object it captures only the wearer's contribution to the closing tendency.

Motion Magnitude: A scalar quantity, M_total, computed per Equation 1 from the reported linear accelerations and angular rates, the rotational term being rendered commensurate with the translational term by an angular-rate weighting constant. It is the single measured quantity from which the adapted detector threshold is derived, and is computed once per cycle irrespective of the number of detections.

Per-State Cooldown Interval: An interval, t_cooldown, during which no repeat actuation is permitted for the same object state tuple, being the combination of class label, directional zone and proximity rank. Its purpose is to prevent repetition of information the wearer already holds, and it is the only suppression interval the escalation criterion may override.

Global Inter-Warning Interval: An interval, t_gap, enforced between any two actuation events of any kind, irrespective of object, class or computed value. Its purpose is distinct: it protects a shared physical output channel from contention and from queueing of stale messages. Because that function would fail if any condition could bypass it, none does, and an escalating threat is therefore re-signalled after the global interval.

Escalation Criterion: A threshold increment, epsilon, applied to the difference between the current threat control value of an object state and the value recorded at its last actuation, per Equation 6. Satisfaction overrides the per-state cooldown alone. The asymmetry between the two suppression intervals is deliberate.

Obstacle Class and Findable Class: Two mutually exclusive sets into which the class gate (500) partitions the recognised inventory. A findable-class detection is prevented from causing walking-mode obstacle-warning actuation irrespective of its computed value, while remaining available to the object-finding function. The gate operates on the label assigned by the detector and cannot suppress a warning arising from an object the detector has itself misclassified into an obstacle class, because in that case no findable-class label exists for it to act upon.

Detector Operating Confidence Threshold: The minimum detector confidence at which a candidate detection is accepted, sigma. The apparatus holds a stationary value sigma_base, a floor value sigma_min below which the threshold may not fall, and an adapted value

sigma_adjusted computed per Equation 2. In the embodiment described the threshold is the parameter adapted. Functionally equivalent detector operating parameters may be substituted.

3. Prior Patents and Publications from Literature

The references below were located during the searches conducted in preparing this disclosure. Publication numbers, titles, authors, volumes, article numbers and digital object identifiers have been checked against the corresponding patent office or publisher record, save where expressly noted. Each comment states what the reference discloses and then what, as described, it does not disclose, in terms matching the present invention. Residual searching still required is set out in Section 3.1, and no determination of novelty or inventive step is asserted here.

A. Prior Patent Literature

S. No.	Patent / Publication No.	Title	Comments
A1	US20230350073A1 (published November 2023; subsequently abandoned)	Edge and Generative AI-Based GPS-Navigated Wearable Device for Blind and Visually Impaired People	CLOSEST APPARATUS-LEVEL ART, distinguished expressly. Discloses a harness-mounted camera with edge-AI identification, distance, position and motion estimation, spatial sound warning of important objects in the path, voice alerts, haptic feedback and a smartphone GPS link; the apparatus platform is accordingly not relied upon as the point of novelty. Distinguished on four grounds: its motion estimation is a detection output applied to the observed scene rather than a measurement fed back to the detector's acceptance threshold; it selects by importance in the path rather than by an ordinal value combining class-calibrated proximity, centrality, apparent size and inertial convergence; as described it contains no temporal output-control stage of any kind; and it does not withhold actuation by class irrespective of computed value. Abandonment does not remove it as prior art, since it was published.
A2	US20240062548A1 / US12327411B2 (Arizona Board of Regents; priority 22 August 2022, granted 23 September 2025)	Converting Spatial Information to Haptic and Auditory Feedback	Claims LiDAR and camera capture, a neural network detecting objects and generating descriptions, a haptic device per direction range, and an auditory output. The three-position haptic arrangement and combined haptic-plus-auditory modality are accordingly not relied upon as novel. As granted it maps object angle to a direction range, disclosing no ranking of simultaneous detections, no temporal scheduling, and no inertial adaptation of the detector.
A3	US9792501B1 (Vasuyantra Corp; Maheriya and	Method and Device for	Discloses a worn computer vision device using head-mounted video sensors, machine vision processing and a haptic transducer band to report

S. No.	Patent / Publication No.	Title	Comments
	Srivastava, granted 17 October 2017)	Visually Impaired Assistance	navigationally important objects. As disclosed it reports detections without an ordinal threat value derived from proximity, centrality, apparent size and inertially measured wearer motion, and without any suppression interval or escalation override governing when a report is permitted.
A 4	US10528815B2 (Vasuyantra Corp, granted 2020)	Method and Device for Visually Impaired Assistance	Continuation of A3 with enhanced object identification, sharing the vision band, haptic band and separate processing hub architecture. As disclosed there is no composite threat control value, no per-class proximity calibration, no actuation cooldown or escalation logic, and no inertial adaptation of detector sensitivity.
A 5	US9801778B2 (A. Mahoney, granted 31 October 2017)	System and Method for Alerting Visually Impaired Users of Nearby Objects	Divides the space ahead into nine collision frusta by stereo reconstruction and drives neck-worn vibro-tactile devices, with a priority decision favouring near over far and centre over lateral. That priority is purely geometric over fixed cells and carries no object class, so it supports neither per-class proximity calibration nor class-dependent withholding. As disclosed there is no inertial measurement, no persistence, cooldown or global interval, and no escalation override.
A 6	US9370459B2 (A. Mahoney, granted 2016)	System and Method for Alerting Visually Impaired Users of Nearby Objects	Earlier member of the A5 family, claiming an imaging device, a processing unit converting images to controller signals, and lanyard-mounted right, left and top vibro-tactile devices. Confirms a three-position body-worn vibrotactile array driven from a camera as long established. As disclosed it does not classify object type, compute a composite value, or apply temporal discipline to output.
A 7	US10299982B2 (D. M. Frankel, granted 2019)	Systems and Methods for Blind and Visually Impaired Person Environment Navigation Assistance	Constructs a three-dimensional representation by multi-view stereo or structure-from-motion and presents it through a touch screen for virtual tactile exploration, being directed to stationary exploration, not to continuous warning during walking. As disclosed there is no real-time ranking of simultaneous detections, no scheduling state machine, and no automatic actuation of a physical transducer.
A 8	US10371544B2 (granted 2019; priority from US provisional 62/501,140 filed 4 May 2017)	Vibrating Haptic Device for the Blind	Discloses a wearable haptic band delivering vibration-encoded directional cues with location tracking between two or more locations, being directed to route guidance from an external positioning estimate rather than obstacle warning from a camera. As disclosed there is no camera-based detection, no object classification, no threat control value and no actuation scheduling.

S. No.	Patent / Publication No.	Title	Comments
A9	US9613505B2 (Toyota Motor Corp; Martinson, Sisbot, Djugash, Oguchi, Takaoka and Nakano, granted 2017)	Object Detection and Localized Extremity Guidance	Discloses an extremity-worn vibration interface and a system generating a trajectory to a tangible object and guiding the extremity along it, being directed to guiding a hand toward an object rather than warning a walking wearer away from one. As disclosed there is no prioritisation between competing objects, no warning-rate discipline, and no voice or spatial-audio modality.
A10	US10690771B2 (granted 2020)	Method and Apparatus for Object Detection Using Human Echolocation for the Visually Impaired	Discloses a parametric transducer array projecting a self-demodulating ultrasonic beam so audible sound returns from distant objects, the user determining position by trained echolocation, not by machine classification. As disclosed there is no class label, no directional zone analysis, no threat control value and no scheduled actuation by the apparatus.
A11	US11792315B2 (granted 2023)	Advanced Eyesight Device for Visually Impaired	Discloses a wearable and cane-associated arrangement combining sensors with variable audible feedback and stereophonic depth estimation, linked to a smartphone application, establishing that depth-derived distance with graded audible feedback is known. As disclosed it describes no composite value ranking simultaneous detections, no four-stage temporal output-control machine, and no inertial adaptation of detector sensitivity.

Table 1: Prior Patent Literature Reviewed by the Inventors

B. Non-Patent Literature

S. No.	Publication / Source	Title	Comments
B1	Tang, J., Zhu, Y., Jiang, G., Xiao, L., Ren, W., Zhou, Y., Gu, Q., Yan, B., Zhang, J., Bi, H., Wu, X., Fan, Z. and Gu, L. (2025), Nature Machine Intelligence, Vol. 7, No. 4, pp. 627–638, DOI 10.1038/s42256-025-01018-6	Human-Centred Design and Fabrication of a Wearable Multimodal Visual Assistance System	Presents a wearable system combining a customised detection algorithm with stretchable artificial skins and triboelectric insoles, evaluated with blind and partially sighted participants. Establishes wearable camera, machine detection and coordinated audio-plus-haptic feedback as known and user-validated, so that combination is not advanced as a basis for novelty. As described it discloses no per-object composite value ranking simultaneous detections, no temporal output-control machine, and no inertial adaptation of the detector threshold.
B2	Pundlik, S., Baliutaviciute, V., Moharrer, M., Bowers, A. R. and Luo, G.	Home-Use Evaluation of a Wearable Collision	Reports a double-masked trial in 31 participants using a chest-mounted camera device with collision-threat estimation and a wrist vibrotactile alert, obstacle contacts falling by approximately 37%. Establishes the

S. No.	Publication / Source	Title	Comments
	(2021), JAMA Ophthalmology, Vol. 139, No. 9, pp. 998–1005, DOI 10.1001/jamaophthalmol.2021.2624	Warning Device for Individuals With Severe Vision Impairments: A Randomized Clinical Trial	chest-mounted camera plus vibrotactile configuration as known and clinically evaluated, so that form factor is not relied upon as inventive. As described it warns without class-dependent withholding, without multi-factor ranking among simultaneous detections, and without an escalation-conditioned override.
B3	Liang, Z., Wu, D. and co-authors (2025), Nature Communications, Vol. 16, Art. 58085, DOI 10.1038/s41467-025-58085-x	A Wearable Obstacle Avoidance Device for Visually Impaired Individuals with Cross-Modal Learning	Reports glasses paired with a smartphone, in which a depth-aided compression module on a field-programmable gate array reduces transmission delay and a cross-modal detection module drives auditory and tactile alerts, being directed to the reliability and latency of the detection chain. As described it discloses no ranking of competing detections, no persistence, cooldown and global-gap scheduling, and no inertial modification of the detector operating point.
B4	Tapu, R., Mocanu, B. and Zaharia, T. (2017), Multimedia Tools and Applications, Vol. 76, pp. 11771–11807, DOI 10.1007/s11042-016-3617-6	A Computer Vision System That Ensure the Autonomous Navigation of Blind People	Describes a mobile obstacle detection pipeline delivering audio alerts through bone-conduction headphones so environmental sound remains audible. Establishes bone-conduction output and deliberate preservation of ambient hearing as long known, and these are accordingly not relied upon as inventive. As described it discloses no composite threat control value, no per-class proximity calibration, and no temporal state machine governing when an alert is permitted.
B5	Shen, J., Chen, Y. and Sawada, H. (2022), Sensors, Vol. 22, No. 12, Art. 4537, DOI 10.3390/s22124537	A Wearable Assistive Device for Blind Pedestrians Using Real-Time Object Detection and Tactile Presentation	Implements a compressed detector on a mobile device driving a shape-memory-alloy tactile display presenting object proximity to the skin. As described it discloses no sensor-responsive threat prioritisation, no scheduling discipline with escalation override, no inertial adaptation of detector sensitivity, and no class-based withholding of actuation.
B6	Wang, W., Jing, B., Yu, X., Sun, Y., Yang, L. and Wang, C. (2024), Sensors, Vol. 24, No. 23, Art. 7621, DOI 10.3390/s24237621	YOLO-OD: Obstacle Detection for Visually Impaired Navigation Assistance	Proposes detector architecture modifications, being a Feature Weighting Block, an Adaptive Bottleneck Block and an Enhanced Feature Attention Head, to raise accuracy in cluttered scenes. As described it addresses neither which of several detections should drive an actuator nor how often actuation may occur, and discloses no wearable actuation hardware or output scheduling.
B7	Bouteraa, Y. (2021), Micromachines, Vol. 12, No. 9, Art. 1082, DOI 10.3390/mi12091082	Design and Development of a Wearable Assistive Device Integrating a	Combines ultrasonic ranging with a fuzzy decision support system and a mixed voice-and-haptic interface. This is the closest reviewed analogue to the present prioritisation stage, since it interposes a decision layer between sensing and output. That layer

S. No.	Publication / Source	Title	Comments
		Fuzzy Decision Support System for Blind and Visually Impaired People	operates on ultrasonic range alone: as described there is no object class, no per-class calibration, no inertial term, and no interval or escalation criterion governing output timing.
B 8	Shah, S. S., Imran, A., Saad-Ur-Rehman, Arif, A., Khan, K., Arsalan, M., Manzoor, S. and Sirewal, G. J. (2026), Automation (MDPI), Vol. 7, No. 2, Art. 41, DOI 10.3390/automation7020041	Vision-Based Smart Wearable Assistive Navigation System Using Deep Learning for Visually Impaired People	Deploys a fine-tuned lightweight detector with a depth camera and threshold-triggered voice feedback reporting obstacles with distance. As described the triggering rule is a fixed distance threshold applied per detection: there is no ranking among simultaneous detections, no persistence or cooldown discipline, no escalation override, and no class-based gate.
B 9	Bai, J., Liu, Z., Lin, Y., Li, Y., Lian, S. and Liu, D. (2019), Electronics, Vol. 8, No. 6, Art. 697, DOI 10.3390/electronics8060697 (preprint arXiv:1904.13037)	Wearable Travel Aid for Environment Perception and Navigation of Visually Impaired People	Performs ground segmentation and walkable-direction search from RGB-D and inertial data with audio feedback. This is the closest reviewed analogue to the inertial aspect of the present invention. The inertial data nonetheless establishes ground-plane orientation for segmentation, a geometric interpretation step; it does not modify the detector's acceptance threshold and contributes no term to any per-object priority value. As described there is no threat control value and no scheduling state machine.

Table 2: Non-Patent Literature Reviewed by the Inventors

Note on citation verification. Every entry carries a full identifier, being a patent publication or grant number, a digital object identifier or both. The identifiers, titles, authors, assignees, volumes, article numbers and page ranges of entries A2 to A11 and B1 to B9 have been checked against the issuing patent office record or the publisher record of record. Entry A1 is recorded as located by the inventors from the published application. Its identifier and subject matter should be re-confirmed against the United States Patent and Trademark Office record, and its full assignee and inventor details obtained, before filing. It is retained notwithstanding, because it is the closest apparatus-level document presently known and the position of Section 4 is stated expressly against it.

3.1 Scope of Prior-Art Searching Still Required

The search reflected in Section 3 was carried out by the inventors and was confined largely to assistive technology for blind and visually impaired persons, and entirely to a single patent jurisdiction. That scope is insufficient in two respects, both of which must be addressed by a professional search before filing.

The first insufficiency is jurisdictional. Every entry in Table 1 is a United States document. European, international, Chinese, Japanese, Korean and Indian filings in the same subject matter have not been searched, and a document from any of those offices may be cited against the application. The professional search must extend to at least the EP, WO, CN, JP and IN collections and to the corresponding classification subclasses covering aids for the blind, tactile signalling systems and scene-specific image recognition.

The second insufficiency is one of technical field. An examiner may rely on a document from an analogous field where it teaches the same technical mechanism, whatever user population that document was written for. Each of the three mechanisms relied upon here has an obvious analogue outside assistive technology. For the threat control value, the search must cover obstacle threat and collision-risk estimation in autonomous mobile robots, advanced driver assistance and forward collision warning, pedestrian safety systems, unmanned aerial vehicle obstacle prioritisation, and industrial proximity alerting. For the temporal output-control process, it must cover alarm suppression and alarm-fatigue management, alert throttling, priority-based notification scheduling, event persistence filtering, hysteresis and debounce logic, and escalation-based bypass of suppression in process control, aviation alerting and driver warning. The third sweep, covering the inertially adaptive detector threshold, reaches motion-adaptive vision confidence, camera-shake and motion-blur compensation, inertially assisted object detection and threshold adaptation, and sensor-fusion gating of detector output in vehicle-mounted and drone-mounted camera systems.

The scope at which each concept identified in Section 10 can safely be claimed is to be settled by that search, and not inferred from the absence of a blind-navigation reference disclosing it.

4. Summary and Background of the Invention (Gap / Novelty)

According to the World Health Organization, at least 2.2 billion people worldwide live with a near or distance vision impairment, of whom an estimated 43 million are blind. Indoor

navigation remains substantially unresolved for this population, because indoor environments present obstacle configurations that develop and disappear over seconds. Conventional aids provide awareness restricted to the point of contact: a long cane reports an object only once the tip reaches it, cannot identify what it is, cannot report its bearing, and cannot distinguish an object to be avoided from one being sought.

4.1 Limitations of Existing Wearable Assistive Installations

Existing electronic installations fall into two categories. Hardware-sensor installations built on ultrasonic or infrared ranging detect presence within a limited range but cannot classify object type, provide object-level directional guidance, or differentiate threat between two simultaneously present objects. Camera-and-neural-network installations classify competently, and Section 3 establishes that wearable edge computing, bone-conduction audio, vibrotactile arrays and combined multi-modal output are all known and, in the case of B2, clinically evaluated. The limitation of that second category lies downstream of detection: the reviewed installations announce on a per-frame or per-object basis, disclose no quantitative basis for deciding which object should drive the output, and disclose no modulation of behaviour according to the measured motion state of the wearer.

A second limitation follows from sensor placement. A body-worn optical sensor moves with the trunk. Motion blur reduces detector confidence and a fixed threshold discards the detection. Within the evaluated dataset 26.1% of otherwise valid detections were lost this way during walking. Permanently lowering the threshold fails, because the same setting admits noise whenever the wearer stands still, which is when the wearer is most likely to be listening carefully.

4.2 Distinctions Over the Closest Reported Systems

Reference A1 is the closest document at the apparatus level and discloses the same component inventory. It is distinguished on the four grounds set out in Table 1, of which the principal two are that its motion estimation is applied to the observed scene and never reaches the detector's operating point, and that it discloses no temporal output-control stage at all. Reference A2, as granted, maps the angle of a detection to one of a plurality of direction ranges and outputs feedback for that range. The mapping is one-to-one from geometry to output and involves neither competition between simultaneous detections nor timing discipline.

Reference A5 requires closer treatment, because it is the one reviewed patent performing a priority decision between simultaneously detected hazards, favouring near over far and centre over lateral. That priority is genuinely analogous to the ordering produced by the proximity and centrality terms here, and prioritisation as such is not treated as novel. The distinction is that the reference orders fixed spatial cells populated by reconstructed points and therefore never possesses an object identity. Without a class label it cannot calibrate a proximity threshold to the physical size of the object, and cannot withhold actuation for an object correctly recognised as navigationally irrelevant. Its scheme is also purely instantaneous: with no persistence requirement, suppression interval, global interval or escalation condition, the ordering is re-applied on every cycle without regard to what was signalled previously.

Reference B7 is the closest analogue to the prioritisation stage in the non-patent literature, interposing a fuzzy decision layer between sensing and output. That layer operates on ultrasonic range alone, carries no class, image geometry or inertial term, and the reference applies no temporal discipline to the resulting output.

4.3 Non-Obviousness

The constituent elements are individually familiar. Wearable edge computing, single-stage detection, inertial measurement units, bone-conduction audio, three-position vibrotactile arrays, threshold-based triggering and suppression timers are each long established, several specifically within assistive technology for blind users. No inventive step is asserted in any one of them, nor in the chest-mounted form factor, nor in the component inventory as a set. The step is asserted in the particular co-operative arrangement, and in three consequences of that arrangement which do not follow from the elements taken separately.

The first consequence is that a single inertial measurement, once placed at two points in one perception-to-actuation loop, couples detection sensitivity to warning priority in a way neither placement produces alone. Considered as an image-stabilisation measure, lowering the detector threshold under motion is an unremarkable compensation. Considered as a priority term, resolving wearer motion onto an object bearing is an unremarkable convergence estimate. Their combination in one loop produces a behaviour that is not the sum of the two: the objects retained by the threshold adaptation are disproportionately objects detected while the wearer is moving, and those same objects receive an elevated convergence term because the wearer is moving

toward them. The adaptation therefore does not recover detections at random. It recovers preferentially the detections the priority computation will rank highest. A practitioner adding motion compensation to a detector, and separately adding a convergence term to a priority rule, would obtain two independent improvements and would not obtain this reinforcement, because the reinforcement is a property of the two stages sharing one measurement within one cycle.

The second consequence is that a temporal output-control stage operating on a ranked threat stream behaves differently from one operating on a raw detection stream, and the difference is measurable. Suppression parameters applied to a raw stream must be set conservatively, because the stream carries no information about which of two suppressed items was the more important. Applied to a ranked stream the same parameters can be aggressive, because the item admitted is by construction the highest-ranked. Section 8.6 records 8 actuation events where a per-frame baseline produced approximately 180 over the same 471 frames, a reduction of 95.6%, with no warning-eligible obstacle missed. A reduction of that magnitude without a missed warning is available only because selection precedes suppression. Reversing the order would suppress before ranking and would discard the highest-ranked object whenever a lower-ranked one arrived first.

The third consequence is that the two suppression intervals must be governed by different override rules, and that this asymmetry is a technical choice. The per-state cooldown exists to prevent repetition of information already held and is properly bypassed when the situation has materially worsened. The global interval exists to protect a shared physical channel from contention and from queueing of stale messages, and would fail in that function if any condition could bypass it. The invention resolves the tension by giving each interval a distinct override rule, so an escalating threat is re-signalled after the global interval. Section 8.6 records that the global interval alone would have permitted approximately 52.7 warnings in the evaluated 79-second window, and that the observed 8 events are 15.2% of that ceiling. The persistence and cooldown stages therefore perform the majority of the reduction, and the global interval performs a different job, which is why it is the interval that must never yield.

4.4 Mapping of Claimed Features Against the Closest Prior Art

The table below maps each element of the principal claim against the four closest references. A tick indicates the element is disclosed; a cross that it is not disclosed as described; a tilde in

parentheses partial or analogous disclosure directed to a different purpose, as explained beneath. The entries record only what was found during the inventors' own review and are not determinations of novelty or inventive steps.

Claimed Element	A1 (edge-AI wearable)	A2 (Arizona)	A5 (Mahoney)	B9 (Bai)
Body-worn imaging sensor with on-device neural object detection over a class inventory	✓	✓	✗	✓
Inertial sensor rigidly co-located with the imaging sensor within a single worn enclosure	(~)	✗	✗	✓
Detector operating confidence threshold modified in dependence upon a measured motion magnitude of the apparatus	✗	✗	✗	✗
Per-object value combining class-calibrated proximity, forward-axis centrality, apparent size and inertial convergence	✗	(~)	(~)	✗
Inertially derived wearer-motion term contributing to that per-object value	✗	✗	✗	✗
Ranked selection of one object per actuation event against an eligibility threshold	(~)	(~)	(~)	(~)
Persistence qualification over successive processing cycles before any warning is issued	✗	✗	✗	✗
Per-object-state suppression interval for repeated warnings	✗	✗	✗	✗
Global minimum interval enforced between successive warnings of any kind	✗	✗	✗	✗
Escalation criterion overriding the per-state interval while retaining the global interval	✗	✗	✗	✗
Class-dependent withholding of actuation for a designated class register	✗	✗	✗	✗
Ranked value drives synchronised voice, vibrotactile and open-ear spatial-audio actuation	(~)	(~)	(~)	(~)

Table 3: Element-by-Element Mapping of the Principal Claim Against the Closest Identified Prior Art

No single reference discloses more than two of the twelve elements. Seven of the twelve, being the third, fifth, seventh, eighth, ninth, tenth and eleventh rows, are not disclosed even partially by any of the four. The tilde entries resolve as follows. In the second row, A1 discloses motion estimation without identifying it as an inertial measurement of the apparatus. In the fourth row, A2 discloses distance and direction only and A5 collision-frustum geometry only. Neither carries apparent size or an inertial term. In the sixth row, all four perform some selection or ordering, A5 by frustum priority, A2 by direction mapping, B9 by walkable-direction search and A1 by

importance in the path, but none against a composite per-object value with an eligibility threshold. In the twelfth row, A1 discloses spatial sound and haptic output, A2 haptic and auditory, A5 vibrotactile alone and B9 audio alone, but in no case is the selection and timing of that output governed by a composite ranked value.

4.5 Technical Effects Achieved by the Invention

Each row below states an effect that is a change in the physical or configured state of the apparatus, or a measurable change in an operating characteristic. None is an item of information presented to the wearer. The individual operations producing these effects, being the setting of a threshold register, the switching of a motor driver stage and the synthesis of a tone, are established capabilities of the constituent hardware. What the invention supplies is the determination of which operation is applied, to which object, at what moment and under what constraint.

Technical Effect	Manner in Which It Is Achieved and Measured
Retention of detections degraded by body motion	The measured motion magnitude lowers the value in the detector threshold register, subject to a floor, and the full threshold is restored when the apparatus is stationary. Motion-induced missed detections fell from 26.1% to 16.1%, being 10.0 percentage points and a relative reduction of 38.3%, at a cost of 2.3 percentage points of additional false detections (Table 11). The effect is a change in the operating behaviour of a physical sensing subsystem.
Reduction in redundant physical actuator activation	Actuation events fell from approximately 180 to 8 over the same 471-frame sequence, a reduction of 95.6%, being one actuation per 58.9 frames against one per 2.6, and 6.1 events per minute against 136.7 (Table 14). Each avoided event is a motor drive stage not switched and an audio channel not occupied.
Allocation of the actuator set to the most collision-relevant object	Useful announcement accuracy remained at 88.5% in fifteen-object scenes against 54.9% for a per-frame baseline, a gap of 33.6 percentage points; accuracy declined 0.74 points per additional object against 3.14 for the baseline, a ratio of 4.2 (Table 10).
Bounded interval between frame capture and actuator activation	The path from capture to actuator drive is 89.2 ms against a 200 ms requirement, leaving 110.8 ms of margin and 55.4% of the budget unused; the unaccelerated path is 228.0 ms and exceeds the requirement by 28.0 ms (Table 16).
Withholding of actuation for navigationally irrelevant classes	The class register prevents actuation for six classes irrespective of computed value. Those classes, when retained during development, produced navigationally irrelevant warnings at 9.4% to 23.1% with a mean of 15.6%; with the register in force no class-inappropriate walking-mode actuation was observed (Table 15).
Preservation of an unoccluded ear canal during audio actuation	Both the spoken channel and the tone channel are routed through the two analogue outputs of an external codec to a left and right open-ear bone-conduction pair, so no occluding element enters the ear canal when

Technical Effect	Manner in Which It Is Achieved and Measured
	both operate. The effect is a permanent property of the signal routing shown in Figure 3.
Graduated encoding of proximity into actuator drive parameters	Vibrotactile pulse frequency rises from 1 Hz at medium proximity to 5 Hz at very close, tone repetition interval falls from 700 ms to 160 ms and pitch rises from 494 Hz to 880 Hz, being a 4.38-fold interval reduction and a 1.78-fold pitch rise across the same buckets (Table 13).
Preservation of warning latency for an escalating threat	Where the escalation criterion is satisfied, the interval to a repeat actuation falls from the 3.0 s per-state cooldown to the 1.5 s global interval, a halving, while the global interval remains in force so channel contention cannot occur (Table 8, Table 14).
Directional resolution of actuation to a bearing	The normalised horizontal coordinate of the bounding-box centre drives both the three-position motor selection and the spoken clock-face bearing. Across 104 reviewed keyframes, 208 zone classifications, being one horizontal and one vertical each, were verified correct against ground truth (Table 11).
Absence of output-channel congestion	No queueing of stale spoken messages was observed. The mechanism is that the 1.5 s global interval exceeds the 45.3 ms speech-initiation stage by a factor of 33, so a new message cannot be issued while a previous one is initialising, and the latest message replaces any stale pending one (Table 16).

Table 4: Technical Effects Produced by the Invention

4.6 Industrial Applicability

The invention is capable of industrial application wherever a body-worn or vehicle-borne sensing unit must convert a continuous stream of detections into a small number of physical warnings, without a display and without saturating the wearer's attention. It requires no component that is not commercially available.

Sector	Manner of Deployment and Technical Benefit
Assistive devices for blind and low-vision users	Deployed as the chest-worn unit of Section 7. The class register keeps household objects out of the walking-mode stream while retaining them for the object-finding function, so one device serves both navigation and retrieval without the two interfering.
Orientation and mobility training services	Deployed as a training aid whose clock-face bearing output matches the convention already taught. The persistence and cooldown parameters are configurable, permitting an instructor to set a conservative warning rate for a novice and a permissive one for an experienced traveller without altering the apparatus.
Low-vision rehabilitation and elderly care	Deployed in residential settings where the obstacle set is largely static and gait is slow. The inertial adaptation contributes least in this regime, motion magnitude being low, while the class register and cooldown contribute most by keeping the output rate low over long periods of wear.
Industrial and warehouse worker proximity safety	Deployed helmet- or vest-mounted where a worker must be warned of moving plant. The global interval is the operative feature: it bounds the rate at which a

Sector	Manner of Deployment and Technical Benefit
	shared alerting channel can be driven, so a worker in a congested aisle receives spaced warnings rather than continuous vibration.
Firefighting and low-visibility emergency response	Deployed in smoke-filled or unlit interiors where the wearer has no usable vision. The measured degradation from 94.2% in bright conditions to 42.8% in darkness, recorded in Section 8.7, bounds applicability directly, and deployment requires substitution of a thermal or active-illumination imaging sensor.
Mobile robotics and delivery robots	Deployed as the warning and prioritisation stage of a small ground robot, the threat control value selecting the single obstacle reported to a remote operator and the scheduling engine bounding message rate over a constrained radio link.
Advanced driver assistance and pedestrian alerting	Deployed as the alert-arbitration stage of a vehicle warning system, the asymmetric override rule being operative: a per-state cooldown prevents repetition of a hazard already announced, while an inviolable global interval prevents two hazards from being announced over one another.
Consumer wearable and augmented-reality notification arbitration	Deployed as an output-throttling stage for a head-worn device with a single audio channel, persistence qualification rejecting transient notifications and the escalation criterion permitting a rising-priority item to break a suppression interval without breaching the channel's minimum spacing.

Table 5: Industrial Applicability of the Invention

4.7 Position Under Section 3(k) of the Patents Act, 1970

The subject matter is a computer-implemented control process embodied in a wearable apparatus. Section 3(k) excludes mathematical methods, business methods, computer programmes per se and algorithms. The Guidelines for Examination of Computer Related Inventions recognise that a computer-related invention is not excluded where the claimed implementation delivers a technical effect or contribution, that the inventive feature must reside in the implementation and not in an abstract algorithm, and that substance decides the question, not claim wording. For each mechanism relied upon, the technical problem, means, operation and effect are stated in Sections 2.1 and 4.5 and measured in Section 8. No protection is sought for any weighting expression as a mathematical method in the abstract.

5. Objective(s) of the Invention

The primary objective is to provide a wearable apparatus in which continuously sensed image and inertial data are converted, within a bounded interval, into a small number of timed physical warnings allocated to the single most collision-relevant object present. The specific objectives are as follows.

(a) To provide a wearable navigation apparatus integrating a wide-angle imaging sensor, a co-located six-axis inertial sensor, a neural accelerator, a bone-conduction transducer pair and a

three-position vibrotactile array within one chest-worn enclosure, in which a threat control value is computed for each detected object and used to select which object, if any, drives the physical guidance actuators.

(b) To provide and validate an inertially adaptive detector confidence threshold that lowers detection sensitivity requirements in proportion to a measured motion magnitude of the apparatus, subject to a floor value, so that detections degraded by motion of the body-worn optical sensor are retained and are thereby capable of causing physical actuation.

(c) To apply the same inertial measurement, resolved onto the angular bearing of each detected object, as a convergence term within that object's threat control value, so that one physical sensor governs both which objects the apparatus detects and how urgently each is signalled.

(d) To provide an announcement scheduling engine applying four-stage temporal control, comprising persistence qualification, a per-object-state suppression interval, a global inter-warning interval that is never bypassed, and an escalation criterion overriding the per-state interval alone, so as to reduce physical actuator activation relative to a per-frame baseline without missing a warning-eligible obstacle.

(e) To make the asymmetry between the two suppression intervals explicit and structural, so a rising threat may be re-signalled ahead of its per-state cooldown while the shared physical output channel remains protected from contention irrespective of any computed value.

(f) To provide a class-based actuation safety gate partitioning the recognised inventory into obstacle classes and findable classes, preventing correctly detected instances of the latter from driving obstacle-warning actuation irrespective of their computed value, while retaining them for a separate object-finding function.

(g) To deliver synchronised multi-modal output comprising spoken guidance, stereo-panned proximity-modulated tones and vibrotactile patterns encoding direction and proximity, with both audio channels routed through an open-ear transducer pair so the ear canal remains unoccluded and ambient hearing is preserved.

(h) To bound the interval between frame capture and actuator activation within a stated responsiveness requirement, and to determine the persistence, suppression and threshold parameters against that requirement.

(i) To validate the apparatus by reduction to practice, the constituent hardware being assembled and interconnected as described and the complete processing path exercised against recorded indoor video with human-verified ground truth, with the scope and limitations of that validation recorded expressly.

6. Working Principle of the Invention (in brief)

The apparatus operates on the principle that indoor navigation assistance, once bounded by object proximity, directional context, wearer motion state and warning-frequency constraints, is a prioritisation and output-scheduling problem. Detection is treated as a solved upstream service. The invention concerns what is done with its output. The operating cycle comprises six stages executed in fixed order on every cycle: inertial adaptation, detection, class gating, position analysis, threat prioritisation, and temporal scheduling followed by actuation, as illustrated in Figure 3.

At the input, a wide-angle camera module captures a frame from the chest-mounted perspective over the camera serial interface, while a co-located six-axis inertial unit reports linear acceleration and angular rate at 100 Hz. Because the inertial unit samples at 100 Hz against a 30 frame-per-second capture rate, approximately 3.33 inertial samples are available per frame, so the motion magnitude is established before the frame it applies to has finished being pre-processed.

The second stage executes detection on the neural accelerator. A primary detector operating on a general inventory is followed by a supplementary detector trained on two indoor-specific classes absent from that inventory. Both run on the accelerator. Non-maximum suppression and all downstream processing run on the host.

The fifth stage computes the threat control value per Equation 3 from the proximity rank, the forward-axis centrality of Equation 4, the normalised apparent size and the inertially derived convergence factor. This is the downstream application of the same inertial measurement. Objects below the eligibility threshold of Equation 5 are suppressed as non-threatening, and the survivors are ranked, with the highest alone passed forward.

The sixth stage is the announcement scheduling engine, whose four filters operate in strict sequence on that single ranked object. The object must first have persisted for at least the

required number of successive cycles, which rejects transient detections. It must then either have cleared its per-state cooldown or satisfy the escalation criterion of Equation 6 against the value recorded at its own last actuation. It must finally clear the global inter-warning interval, which no condition bypasses. Only then are the actuators driven.

Actuation is simultaneous across three modalities. Spoken guidance is delivered through the open-ear bone-conduction pair, with a name-confidence threshold substituting a generic label for a low-confidence obstacle so the warning survives an uncertain identification. Synthesised tones are delivered through the same pair, stereo-panned to the object's horizontal position with repetition rate and pitch rising with proximity. Vibrotactile patterns are delivered through the three-position array, direction encoded by selective activation and proximity by pulse-frequency modulation. The graduated design across three proximity buckets is what permits the wearer to distinguish an approaching obstacle from a distant one without any spoken content at all.

6.1 Statement of the Invention as Distinct from One Example Implementation

For the avoidance of doubt, and to assist in assessing claim scope, the following separation is drawn between what is put forward as the invention and what is described as one example implementation used for reduction to practice.

What Is Being Invented (Claimed)	One Example Implementation (Not Limiting)
A wearable apparatus in which an imaging sensor, a rigidly co-located inertial sensor, a processor and at least one physical guidance actuator are arranged in a control path from sensing to actuation.	A chest-worn enclosure of 120 mm by 80 mm by 30 mm and approximately 180 g with adjustable elastic straps.
A body-worn imaging sensor of wide field of view supplying frames to a detector at a rate sufficient for continuous operation.	A camera module with an OV5647 sensor and a 160-degree lens over the camera serial interface at 30 frames per second. Any imaging sensor of comparable field of view and frame rate may be substituted.
An inertial sensor rigidly co-located with the imaging sensor and reporting linear acceleration and angular rate.	A six-axis MPU6050 unit on a two-wire bus at address 0x68, sampling at 100 Hz. Any six-axis unit of comparable rate may be substituted.
A processor and neural accelerator executing an object detector whose acceptance threshold is settable at run time.	A quantised single-stage detector compiled for a USB-attached 4 TOPS accelerator at 640 by 640 input resolution, hosted by an ARM Cortex-A72 compute module at 1.5 GHz with 4 GB of memory. Any detector and accelerator producing class-labelled bounding boxes with confidence scores may be substituted.

What Is Being Invented (Claimed)	One Example Implementation (Not Limiting)
A supplementary detector supplying class labels for domain-specific objects absent from the primary inventory.	A smaller model trained by transfer learning on annotated indoor data for two additional classes, executed in sequence after the primary detector on each frame.
An inertial adaptation unit modifying a detector operating parameter in dependence upon a measured motion magnitude, subject to a floor value.	The parameter modified was the detector confidence threshold, using the constants of Equation 2 and the ranges of Table 8. Functionally equivalent detector operating parameters may be substituted, subject to support in the specification.
A threat prioritisation unit reducing each surviving detection to one ordinal value derived from at least a proximity measure, a forward-axis position measure and an inertially derived motion term.	The four-term weighted combination of Equation 3 with the weights of Table 8, selected by exhaustive grid search over 1,771 admissible combinations at 0.05 resolution. Other monotonic combining functions may be substituted.
An eligibility comparator below which no object is admitted for actuation.	A scalar threshold of 0.30, calibrated as described in Section 8.8.
An announcement scheduling engine comprising a persistence gate, a per-state suppression interval, a global inter-warning interval and an escalation comparator overriding the former alone.	Four stages with the values of Table 8, being a two-frame persistence requirement, a 3.0 s per-state cooldown, a 1.5 s global interval and an escalation increment of 0.1. The number of stages is fixed by the invention; the values are configurable.
A class register partitioning the recognised inventory such that designated classes cannot cause obstacle-warning actuation.	Fifteen obstacle classes and six findable classes, the split determined by measured rates of navigationally irrelevant warnings as recorded in Table 15.
At least one physical guidance actuator driven in dependence upon the selected object, with proximity encoded in a drive parameter.	A left and right open-ear bone-conduction pair driven from an external stereo codec, three eccentric-rotating-mass motors switched by discrete transistor stages from three separate output lines each with a flyback diode, and a synthesised stereo tone channel sharing the transducer pair.

Table 6: Separation Between the Claimed Invention and the Example Implementation

6.2 Statement Regarding Hardware Implementation

The invention comprises a wearable warning apparatus configured to observe an environment through an imaging sensor and an inertial sensor and to drive physical guidance transducers automatically through the setting of a detector threshold register, the selection of one object from a ranked set, and the timed switching of motor driver stages and audio output channels. It is constituted from commercially available computing, sensing and transducer elements, comprising a processor and memory, a neural accelerator, an imaging sensor, an inertial sensor, a microphone, an audio codec and amplifier, bone-conduction transducers, eccentric-rotating-mass motors with discrete driver stages, and a lithium-polymer cell with charge management and boost conversion.

6.3 Technical Contribution

- **A dual-stage application of one inertial measurement.** A single motion magnitude, computed once per cycle, sets the acceptance threshold of the detector and, resolved onto the bearing of each surviving object, contributes a term to that object's priority value.
- **A ranked threat-to-actuation control path.** Every surviving detection is reduced to one ordinal value combining class-calibrated proximity, image-derived forward-axis centrality, uncalibrated apparent size and inertially derived convergence, and exactly one ranked object is admitted per event. The actuator set is thereby allocated to one object.
- **A temporal stage operating on a ranked stream.** Suppression parameters can be set aggressively precisely because the item they admit has already been selected as the highest-ranked, which is why a 95.6% reduction in actuation events was obtained without a missed warning within the evaluated sequence.
- **Two suppression intervals with distinct override rules.** The per-state cooldown may be overridden by a rising threat because its function is to avoid repeating known information. The global interval may never be overridden because its function is to protect a shared physical channel. The asymmetry is structural and no computed value alters it.
- **Class-conditioned withholding of actuation.** A register of designated classes prevents actuation irrespective of computed value, so a correctly recognised but navigationally irrelevant object cannot occupy the actuator set, while remaining available to the object-finding function of the same apparatus.
- **Parameters determined against the timing budget.** The persistence requirement, suppression intervals and eligibility threshold were selected against a stated responsiveness requirement, and Section 8.8 records that the persistence value is bounded above by that requirement rather than by its false-alarm behaviour.
- **Graduated physical encoding.** Proximity is encoded in the drive parameters themselves, as pulse frequency, tone repetition interval and pitch, so severity is conveyed by the physical character of the actuation as well as by its spoken content.
- **Independence from any particular detector.** The architecture operates without alteration where the detector is any component producing class-labelled bounding boxes with confidence scores and a settable acceptance threshold, the detector being defined by its position in the control path and the form of the output it delivers.

Each of the foregoing is a property of the arrangement and of the constraints imposed upon its operation, and none is a property of any computational technique executed within it, and each is expressed in Section 10 as a structural or operational limitation upon the apparatus claimed.

6.4 Brief Description of the Drawings

Figure 1 is a system architecture diagram showing the chest-worn unit with its sensing elements and processing core, the three control processes, the multi-modal output controller and its transducers, and the companion smartphone application.

Figure 2 is a layout of the chest-mounted unit showing the placement of the imaging sensor, the inertial sensor, the microphone and the three vibrotactile motor positions, together with the enclosure dimensions.

Figure 3 is a control-process flowchart of the threat prioritisation unit and the announcement scheduling engine, showing each decision gate in sequence with the disposition of an object failing each, and showing expressly that the escalation test bypasses the per-state cooldown alone.

Figure 4 presents the behaviour of the threat control value: its distribution by proximity and centrality category against the eligibility threshold, useful announcement accuracy against scene complexity for the apparatus and a per-frame baseline, directional zone accuracy by zone, and the distribution of total processing path latency against the responsiveness requirement.

Figure 5 presents detection and position analysis performance: per-class detection accuracy, proximity-bucket accuracy, the effect of inertial adaptation on frame-level detector confidence against the acceptance threshold, and the comparison between candidate detector models.

Figure 6 presents the output discipline of the scheduling engine: actuation counts against a per-frame baseline, the timeline of actuation events, the vibrotactile pulse patterns by proximity bucket, and the tone interval and pitch parameters by bucket.

Reference numerals are keyed to the drawings below. The 100 series denotes sensing elements, the 200 series processing elements, the 300 series output transducers, the 400 series the inertial adaptation stage, the 500 series the class gate, the 600 series position analysis, the 700 series threat prioritisation, the 800 series output scheduling, and the 900 series the resulting physical effect and ancillary elements.

Numeral	Element	Numeral	Element
100	Sensor and input layer of the wearable unit	520	Findable-class register
101	Wide-angle camera module	600	Position analysis module

102	Six-axis inertial measurement unit	610	Clock-face bearing unit
103	MEMS microphone for voice commands	700	Threat prioritisation unit (PWTPA)
200	Edge-computing processing core	710	Eligibility comparator
201	Neural accelerator	800	Announcement scheduling engine (AASE)
202	Host controller and compute module	810	Persistence counter store
203	Supplementary indoor detection model	820	Per-state cooldown timer
210	Primary object detection model	830	Global inter-warning interval timer
300	Multi-modal output layer	840	Escalation comparator
301	Open-ear bone-conduction transducer pair	850	Name-confidence comparator
302	Three-position vibrotactile motor array	900	Modified physical actuator state
303	Spatial audio synthesiser	910	External audio codec and stereo amplifier
400	Inertial adaptation unit	920	Discrete transistor motor driver stages
410	Detector threshold register	930	Power management and cell
500	Class-based actuation safety gate (CASG)	950	Object memory store
510	Obstacle-class register	960	Companion smartphone application

Table 7: Reference Numerals Keyed to Figures 1 to 4

Figure 1 carries numerals 100 to 103, 200 to 202, 300 to 303, 700, 800 and 960. Figure 2 carries 101, 102, 103, 201, 202, 301 and 302 against the physical elements it shows. Figure 3 carries 400, 500, 600, 700, 710, 800, 810 to 840, 301, 302 and 303 against the decision stages and outputs. Figure 4 carries numeral 700 against the threat control value it plots. Where a numeral denotes a group of elements shown together, the constituent sub-numerals apply to the elements within that group.

7. Description of the Invention in Detail

[0001] Scope of Implementation. The implementation described uses a quantised single-stage detector compiled for a USB-attached neural accelerator, an ARM compute module host, a six-axis inertial unit, open-ear bone-conduction transducers and eccentric-rotating-mass motors as the platform for reduction to practice. These are representative choices and not mandatory elements. The apparatus extends to any functionally equivalent alternatives, including any detector producing class-labelled bounding boxes with confidence scores and a settable acceptance threshold, any host processor of comparable throughput, any six-axis unit of comparable sampling rate, and any transducer arrangement delivering direction-encoded and

proximity-encoded physical output without occluding the ear canal. Numerical values are disclosed as preferred values within the ranges of Table 8.

[0003] Interconnection. All sensors and actuators interface with the host controller (202), as shown in Figure 3. The camera (101) connects over the camera serial interface, the inertial unit (102) over the two-wire bus at address 0x68, and the microphone (103) over the digital audio input. The accelerator (201) is a USB-attached co-processor and hosts no sensor. Image data reaches it only via the host, so camera and accelerator are not directly connected. The host provides no analogue audio output, so audio is delivered through an external codec with stereo amplifier (910), whose left and right outputs drive the open-ear pair (301) used for both the spoken and tone channels. The three motors (302) are switched independently by three discrete transistor stages (920) from three separate output lines, each stage with a flyback diode across its motor. No dedicated haptic-driver integrated circuit is used, which is what keeps the three positions independently addressable and permits pulse patterns to be generated in firmware by timed switching.

[0005] Inertial Adaptation Unit. The unit (400) computes a motion magnitude and writes the resulting acceptance threshold into the register (410) before detection is executed on the current frame.

Equation 1: Motion Magnitude

$$M_{total} = \sqrt{a_x^2 + a_y^2 + (a_z - g)^2} + \lambda \cdot \sqrt{\omega_x^2 + \omega_y^2 + \omega_z^2}$$

where a_x, a_y and a_z are the measured linear accelerations, g is gravitational acceleration, ω_x, ω_y and ω_z are the measured angular rates, and $\lambda = 0.25$ is the angular-rate weighting constant within the range of Table 8. Subtracting g from the vertical axis removes the static gravitational component so a stationary apparatus yields a magnitude near zero. The constant λ renders the rotational term commensurate with the translational term, which is necessary because the two carry different physical units while a rotation of the trunk degrades the captured image as effectively as a translation.

Equation 2: Adapted Detector Operating Threshold

$$\sigma_{adjusted} = \max(\sigma_{min}, \sigma_{base} - \eta \cdot \min(M_{total}, M_{max}))$$

where sigma_base = 0.6 is the stationary threshold, sigma_min = 0.35 the floor value, eta = 0.15 the adaptation gain, and M_max = 2.5 m/s² a saturation limit set at the 95th percentile of measured walking-pace accelerometer magnitudes. The value written to the register (410) is bounded below by sigma_min and above by sigma_base, and the saturation limit prevents a single impulsive event such as a footfall from driving the threshold to its floor. Because the retained detections are eligible to drive the actuators, this update alters the physical warning behaviour of the apparatus and not merely its internal record.

[0007] Class-Based Actuation Safety Gate. The gate (500) partitions every surviving detection by reference to two registers. The obstacle register (510) holds fifteen classes: person, chair, couch, bed, dining table, bench, toilet, sink, refrigerator, television, potted plant, suitcase, backpack, door and dustbin. The findable register (520) holds six: bottle, cup, laptop, cell phone, book and toothbrush. A detection whose label appears in the findable register is removed from the walking-mode stream at this point, before any threat value is computed for it, and passed instead to the object-finding function. Placing the gate upstream of prioritisation ensures a findable-class object cannot occupy the single ranked actuation slot even transiently. The rationale is physical: a small object at close range occupies enough of the frame to satisfy any area-based proximity criterion while presenting no collision risk, so a cup correctly identified as a cup would otherwise generate an obstacle warning. It is expressly recorded that the gate operates on the label assigned by the detector: it removes irrelevant warnings arising from correctly classified findable objects, and cannot suppress a warning arising from an object the detector has misclassified into an obstacle class, because in that case no findable label exists for it to act upon.

[0008] Position Analysis Module. For each remaining detection the module (600) classifies direction using a three-by-three grid zone, being left, centre or right by top, middle or bottom, from the bounding-box centre coordinates normalised to the interval from 0 to 1, and estimates proximity using per-class area thresholds calibrated for each of the 21 target classes. A person is classified as very close at 35% or more of frame area, while a bottle is very close at 5% or more, which accounts for the inherent size difference. The clock-face unit (610) maps the same normalised horizontal coordinate to bearings in the 10-11-12-1-2 o'clock range for spoken output. Deriving the coarse zone and the finer bearing from one coordinate guarantees that the centrality term and the bearing spoken to the wearer can never disagree.

[0009] Threat Prioritisation Unit. The unit (700) computes a threat control value for each detection surviving the gate.

Equation 3: Threat Control Value

$$T(o) = \text{alpha} \cdot P(o) + \text{beta} \cdot C(o) + \text{gamma} \cdot S(o) + \text{delta} \cdot M(o)$$

where P(o) is the class-calibrated proximity rank taking 1.0 for very close descending to 0.0 for far, C(o) is the forward-axis centrality of Equation 4, S(o) is the normalised bounding-box area, M(o) is the inertially derived convergence factor, and the weights take alpha = 0.40, beta = 0.25, gamma = 0.15 and delta = 0.20 within the ranges of Table 8, subject to alpha + beta + gamma + delta = 1.0.

Equation 4: Forward-Axis Centrality

$$C(o) = 1 - 2 \cdot | x_center - 0.5 |$$

where x_center is the horizontal coordinate of the bounding-box centre normalised to the interval from 0 to 1. The expression returns 1.0 for an object on the horizontal centre line and 0.0 at either edge. The forward axis is expressly a nominal axis represented by the horizontal centre of the captured image, and not an independently reconstructed walkable path.

[0010] Distinction Between the Four Terms. P(o) and S(o) are both derived from bounding-box area but serve distinct functions and are not interchangeable. P(o) is class-calibrated and reports how near the object is. S(o) is uncalibrated and reports how much of the field of view it occupies irrespective of identity. They diverge in practice: a small object very near the wearer yields high P(o) with low S(o), a large object at a distance yields the converse, and an object both near and large is ranked above either. M(o) is distinct from C(o) in the same way: C(o) depends only on image bearing, whereas M(o) additionally depends on the instantaneous measured motion state, so the same centrally located object retains the same C(o) while receiving a different M(o) according to whether the wearer is stationary, approaching or receding. It is because the four terms are independent in this sense that the weighted combination orders objects any single term would rank equally.

[0011] Derivation of the Convergence Factor. The forward acceleration and yaw rate reported by the unit (102) characterise the instantaneous inertial state of the apparatus. The angular bearing of each detection is obtained from its bounding-box centre coordinate through the same

normalised horizontal mapping used for the zone and clock-face outputs, and M(o) is the component of the wearer's own motion directed toward that bearing, quantised to approaching at 1.0, stationary-near at 0.5 or receding at 0.0. Because the bearing term differs between objects, M(o) differs between objects in the same frame, and a centrally located obstacle receives a higher motion term than a peripheral one while the wearer walks forward. It is expressly recorded that M(o) is a wearer-motion-derived convergence indicator for static obstacles, and it does not measure relative object velocity, since an exact closing rate would additionally require calibrated velocity integration, orientation compensation and range information. For an independently moving object it captures only the wearer's contribution to the closing tendency. Resolving the object's own velocity would require combination with frame-to-frame image displacement, which the present implementation does not do.

[0012] Eligibility Comparator. The comparator (710) applies a scalar eligibility threshold.

Equation 5: Eligibility Condition

$T(o) \geq \text{tau}$, where $\text{tau} = 0.30$ within the range of Table 8

Objects below tau are suppressed as non-threatening and cause no actuation of any kind. The survivors are ranked by T(o) and the highest alone passes to the scheduling engine. The threshold performs a different function from the ranking: the ranking decides which object would be signalled, and the threshold decides whether any object is signalled at all, so a frame containing only distant peripheral objects produces silence.

[0013] Announcement Scheduling Engine. The engine (800) operates as a state machine with per-object-state tracking. Each tracked state is a tuple of class label, directional zone and proximity rank, carrying a persistence count in the store (810), the timestamp of its last actuation against the cooldown timer (820), and the threat control value recorded at that actuation for the escalation comparator (840). The four stages operate in strict sequence on the single ranked object supplied by the comparator (710), as shown in Figure 3.

At stage one the persistence counter (810) is incremented for the current object state, and the object is rejected if the count is below n_persist = 2 successive cycles. At stage two the cooldown timer (820) checks elapsed time since the last actuation for the same state tuple against t_cooldown = 3.0 s, and withholds the object if that interval has not elapsed, unless stage four

applies. At stage three the global timer (830) checks elapsed time since any actuation against t_gap = 1.5 s, and withholds the object if that interval has not elapsed. At stage four, reached only where stage two has withheld the object, the comparator (840) applies the following condition.

Equation 6: Escalation Condition

$T(o)_{current} > T(o)_{last} + \varepsilon$, where $\varepsilon = 0.1$ within the range of Table 8

and $T(o)_{last}$ is the value recorded at the last actuation for the same object state. Where satisfied, the per-state cooldown of stage two is bypassed and the object proceeds to stage three. The global interval of stage three is not bypassed and continues to apply in every case. The consequence is that an escalating threat is re-signalled after t_gap rather than t_cooldown, halving the interval from 3.0 s to 1.5 s, and never after zero. The asymmetry is deliberate: the per-state cooldown exists to prevent repetition of information already held and may therefore be bypassed when the threat has materially worsened, whereas the global interval exists to protect a shared physical output channel from contention and from queueing of stale warnings, and would fail in that function if any condition could bypass it.

[0014] Name-Confidence Substitution. The comparator (850) applies a name-confidence threshold of sigma_name = 0.8 to the detection confidence of an obstacle admitted for actuation. Where the confidence falls below that value, a generic label is substituted for the specific class label in the spoken output while the warning itself, and the accompanying vibrotactile and tone output, are preserved unchanged. The purpose is to prevent a confident-sounding misidentification without withholding the warning, since the wearer's first need is to know that something is ahead, and only then what it is.

[0015] Actuation Layer. The selected object drives three modalities simultaneously, producing the modified actuator state (900). The bone-conduction pair (301) delivers spoken guidance through vibrations transmitted via the cheekbone, leaving the ear canal open for ambient sound, which is a safety property and not a comfort preference. The pair comprises a left and a right open-ear transducer driven from the two channels of the external codec (910), so stereo panning is available. The synthesiser (303) generates stereo-panned tones through that same open-ear pair, with pan position mapped to the normalised horizontal coordinate and repetition interval and pitch modulated by proximity rank from 700 ms at 494 Hz for medium to 160 ms at 880 Hz

for very close. The array (302) comprises three motors at left, centre and right, with direction encoded by selective activation and proximity by pulse-frequency modulation from 1 Hz at medium to 5 Hz at very close. Spoken output uses a non-blocking threaded engine in which the latest message replaces any stale pending message, so a warning about an object already passed cannot be queued behind the current one. The term spatial audio denotes synthesised tones used to encode position. No acoustic ranging or echo processing is performed.

[0017] Physical Construction and Power. The apparatus operates on a compute module with an ARM Cortex-A72 processor at 1.5 GHz and 4 GB of memory, powered by a 5000 mAh lithium-polymer cell with charge management and boost conversion to 5 V, in the power block (930). All components are housed in a chest-mounted enclosure of 120 mm by 80 mm by 30 mm, being 288 cm^3, at approximately 180 g, corresponding to a mean packing density of 0.625 g/cm^3, with adjustable elastic straps. Continuous operating time is at 6 to 8 hours from a component current budget. That figure has been confirmed by measurement of the assembled unit under sustained detection load, and achievable runtime depends on cell voltage, boost-converter efficiency and the duty cycle of the accelerator and motors. During assembly, routing the motor supply path separately from the logic supply reduced unstable behaviour observed when all three motors were switched simultaneously at the 5 Hz pulse rate.

[0018] Best Method of Performing the Invention. The best method known to the inventors at the date of this disclosure is the arrangement of paragraphs [0002] to [0017], namely: a 160-degree imaging sensor (101) at 30 frames per second and a six-axis inertial unit (102) at 100 Hz rigidly co-located within one chest-worn enclosure; an inertial adaptation unit (400) applying Equation 1 with lambda = 0.25 and Equation 2 with sigma_base = 0.6, sigma_min = 0.35, eta = 0.15 and M_max = 2.5 m/s^2; a quantised primary detector (210) on a USB accelerator (201) followed by a two-class supplementary model (203); a class gate (500) with fifteen obstacle and six findable classes; a position analysis module (600) with per-class area thresholds and a clock-face unit (610); a threat prioritisation unit (700) applying Equation 3 with alpha = 0.40, beta = 0.25, gamma = 0.15 and delta = 0.20 and tau = 0.30; a scheduling engine (800) with n_persist = 2, t_cooldown = 3.0 s, t_gap = 1.5 s and epsilon = 0.1, in which the comparator (840) bypasses the timer (820) alone; a name-confidence threshold sigma_name = 0.8; and simultaneous actuation through an open-ear pair (301), a three-position array (302) at 1 to 5 Hz and a tone synthesiser (303) at 700 to 160 ms and 494 to 880 Hz. The values stated as defaults in

Table 8 are those found to give the best balance between coverage of warning-eligible obstacles and restraint of actuation rate within the validation dataset and the 200 ms responsiveness requirement.

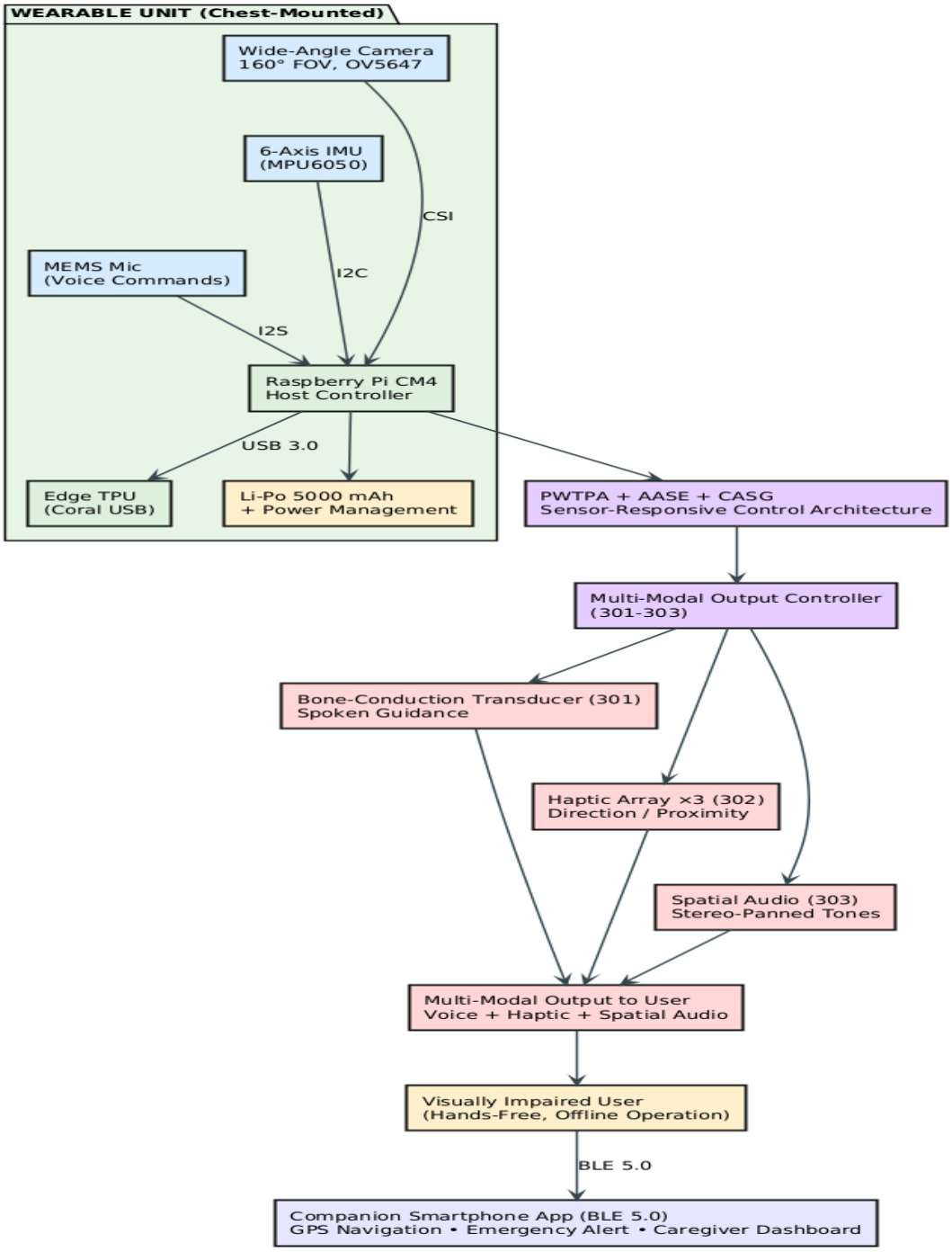


Figure 1: System Architecture of the Wearable Assistive Navigation Apparatus

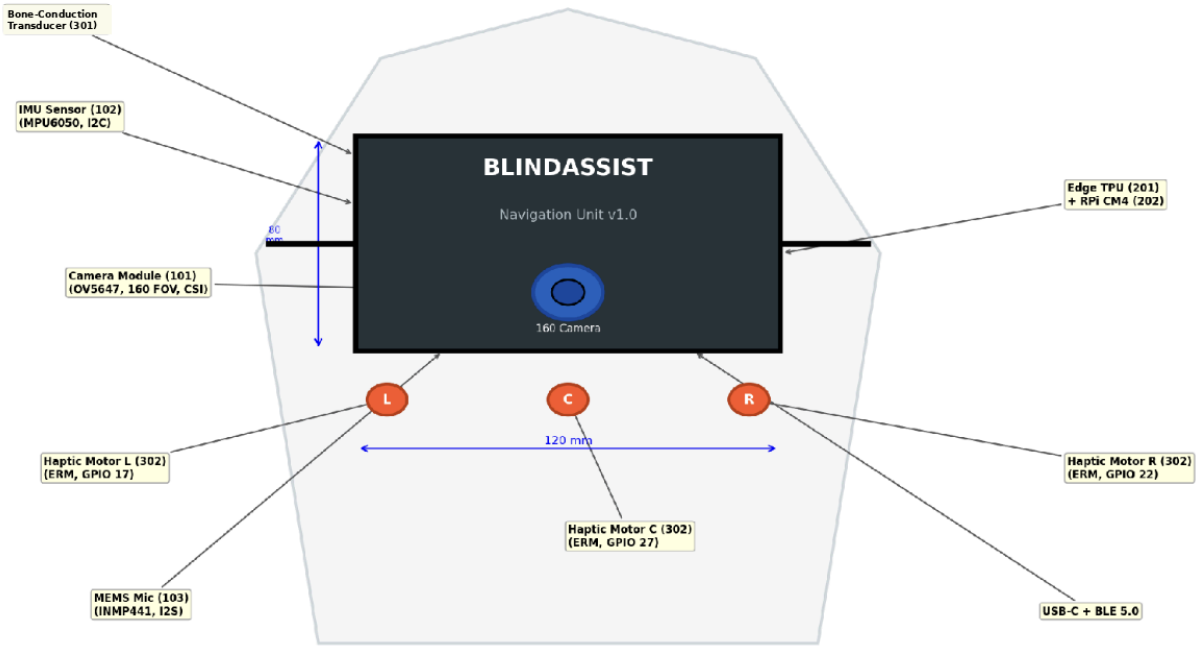


Figure 2: Layout of the Chest-Mounted Wearable Unit Showing Sensor, Actuator and Enclosure Geometry

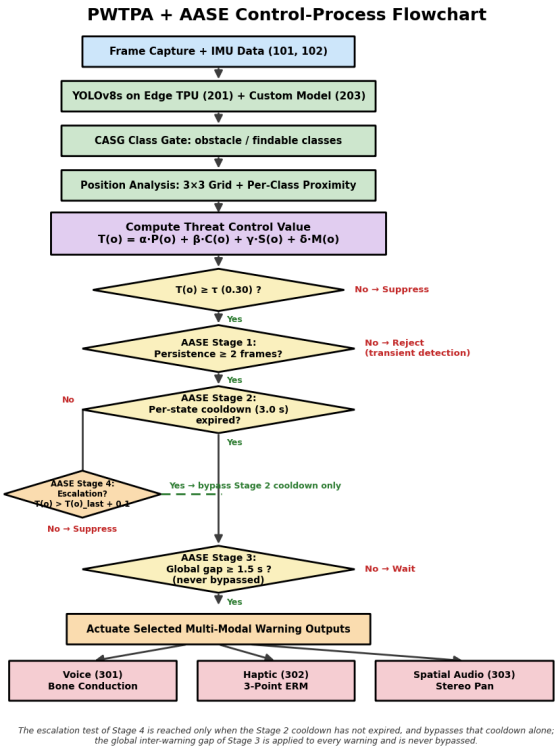


Figure 3: Control-Process Flowchart of the Threat Prioritisation Unit and the Announcement Scheduling Engine with Decision Gates

8. Reduction to Practice and Validation Measurements

The measurements below demonstrate reduction to practice and quantify the effects of Section 4.5. They are engineering validation results obtained during development, not the results of a clinical study or user trial. No visually impaired participants were involved and no claim of clinical benefit is made. All figures apply to the dataset of Section 8.3 only, and the limitations are set out in Section 8.10.

8.1 Validation Setup

Validation proceeded through three layers: 78 automated unit tests across seven processing modules, being position analysis, the eligibility and scheduling logic, speech output, voice command recognition, the monitoring interface, clock-face bearing derivation and object memory; recorded-video evaluation using seven smartphone-captured indoor clips comprising approximately 2,400 annotated frames, a mean of 343 per clip, processed through a reproducible harness with video-time clock synchronisation so every reported interval is measured against one clock; and human keyframe review, in which every actuation saved an annotated frame subsequently verified against ground truth.

The responsiveness requirement against which timing results are judged is 200 ms from frame capture to actuator drive. That figure is the interval within which a warning remains actionable for a walking wearer: at an indoor pace of approximately 1.2 m/s, 200 ms corresponds to roughly 0.24 m of travel, which is within the margin available to arrest a step. It is stated before the results so the results may be judged against it.

8.2 Configuration Parameters

The defaults were established by the calibration and ablation of Section 8.8. Each is stated with the range within which it may be varied without departing from the invention, and with the physical actuation behaviour it controls, since no parameter of the apparatus is purely internal.

Parameter	Component	Default	Permitted Range	Function in Actuation Control
Angular-rate weighting lambda	Inertial adaptation (400)	0.25	0.10 – 0.50	Sets how strongly trunk rotation, as against translation, contributes to the motion magnitude that lowers the detector threshold

Parameter	Component	Default	Permitted Range	Function in Actuation Control
Stationary threshold sigma_base	Threshold register (410)	0.60	0.50 – 0.70	Acceptance threshold at rest; sets the noise floor for actuation while the wearer stands still
Threshold floor sigma_min	Threshold register (410)	0.35	0.25 – 0.45	Lowest threshold permitted under motion; bounds how many marginal detections may drive actuation
Adaptation gain eta	Inertial adaptation (400)	0.15	0.05 – 0.30	Rate at which measured motion lowers the acceptance threshold
Saturation limit M_max	Inertial adaptation (400)	2.5 m/s²	1.5 – 4.0 m/s²	Caps motion magnitude so a single footfall impulse cannot drive the threshold to its floor
Proximity weight alpha	Threat unit (700)	0.40	0.25 – 0.55	Weight of class-calibrated proximity in selecting which object drives the actuators
Centrality weight beta	Threat unit (700)	0.25	0.10 – 0.40	Weight of position relative to the nominal forward axis
Apparent-size weight gamma	Threat unit (700)	0.15	0.05 – 0.30	Weight of uncalibrated visual footprint in the field of view
Convergence weight delta	Threat unit (700)	0.20	0.10 – 0.35	Weight of the inertially derived wearer-motion term
Eligibility threshold tau	Comparator (710)	0.30	0.20 – 0.40	Value below which no object causes any actuation at all
Persistence requirement n_persist	Counter store (810)	2 cycles	1 – 3 cycles	Successive cycles a detection must survive before it may drive the actuators
Per-state cooldown t_cooldown	Cooldown timer (820)	3.0 s	1.5 – 5.0 s	Interval during which a repeat warning for the same object state is withheld
Global interval t_gap	Global timer (830)	1.5 s	1.0 – 3.0 s	Minimum interval between any two actuation events; never bypassed
Escalation increment epsilon	Escalation comparator (840)	0.10	0.05 – 0.20	Rise in threat value permitting the per-state cooldown, and only that cooldown, to be bypassed

Parameter	Component	Default	Permitted Range	Function in Actuation Control
Name-confidence threshold sigma_name	Comparator (850)	0.80	0.70 – 0.90	Confidence below which a generic label replaces the class label in speech while the warning is preserved
Object memory decay interval	Memory store (950)	30 s	15 – 60 s	Period for which last-seen guidance remains available after an object leaves view

Table 8: Configuration Parameters and Their Function in Actuation Control

8.3 Validation Dataset and Identifiers

Data Category	Parameters	Identifiers
Camera input	Wide-angle sensor, 160-degree field of view, camera serial interface, 30 frames per second	CAM-1
Detection models	Primary 80-class single-stage detector, quantised at 640 by 640; supplementary two-class indoor model for door and dustbin	DET-1, DET-2
Inertial sensor	Six-axis unit, two-wire bus at address 0x68, 100 Hz sampling	IMU-1
Threat parameters	alpha = 0.40, beta = 0.25, gamma = 0.15, delta = 0.20, tau = 0.30	PWA-1 to PWA-5
Scheduling parameters	n_persist = 2 cycles, t_cooldown = 3.0 s, t_gap = 1.5 s, epsilon = 0.10	AAS-1 to AAS-4
Class registers	15 obstacle classes, 6 findable classes, 21 in total	CLS-1, CLS-2
Output actuators	Open-ear bone-conduction pair, three eccentric-rotating-mass motors, stereo tone synthesiser	OUT-1 to OUT-3
Video dataset	7 indoor clips, approximately 2,400 annotated frames, mean 343 per clip; warning-discipline sequence of 471 frames spanning 79 s of video time	VID-1 to VID-7
Keyframe review set	104 keyframes, each carrying one horizontal and one vertical zone classification, 208 in total	KEY-1
Unit test suite	78 automated tests across seven processing modules	UT-1

Table 9: Validation Dataset Parameters and Corresponding Identifiers

The 471-frame sequence spans 79 seconds of video time, an effective processed rate of 5.96 frames per second. That figure governs the persistence requirement: two cycles correspond to about 0.34 s of elapsed time in this sequence, not the 0.067 s that two frames represent at the 30 frame-per-second capture rate. Persistence therefore costs a third of a second of delay in the evaluated configuration, and raising the processed rate toward the capture rate reduces that cost proportionally.

8.4 Threat Control Value Behaviour and Announcement Accuracy

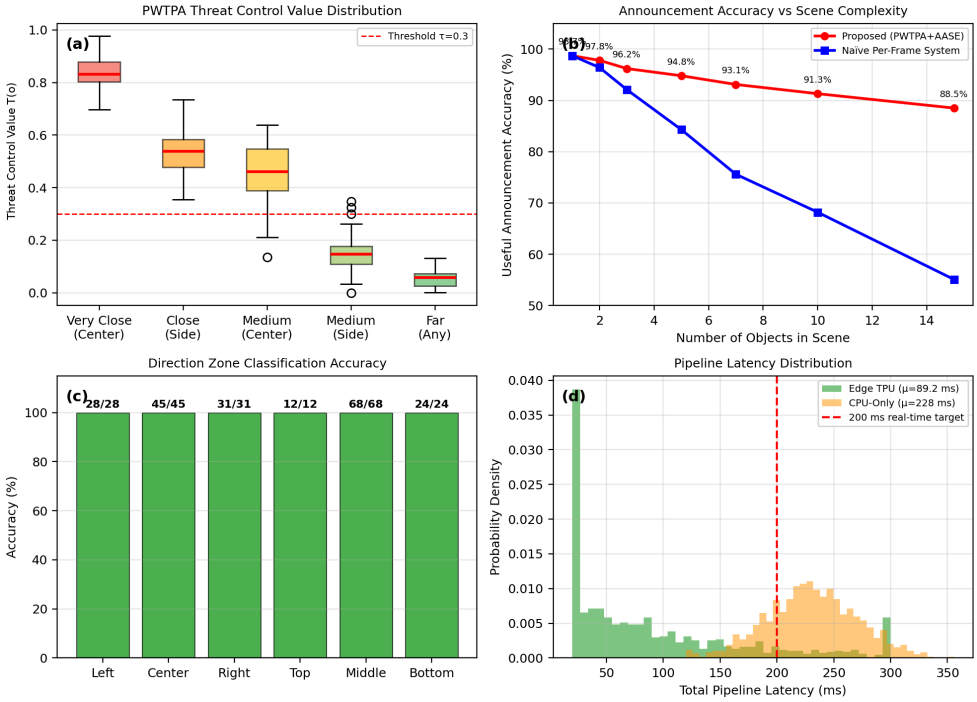


Figure 4: Threat Control Value Distribution, Useful Announcement Accuracy Against Scene Complexity, Directional Zone Accuracy, and Total Processing Path Latency

Panel (a) plots the distribution of the threat control value across five combined proximity-and-centrality categories against the eligibility threshold of 0.30. The three categories that ought to produce actuation, being very close at centre, close at side and medium at centre, have their entire interquartile ranges above the threshold with medians at approximately 0.84, 0.54 and 0.46, while the two that ought not, being medium at side and far at any bearing, have medians at approximately 0.15 and 0.06. The separation arises from the interaction of the proximity and centrality terms: medium-centre and medium-side share the same proximity rank and the same weight alpha, and are separated only by the centrality term and the associated difference in the convergence term. This establishes the eligibility threshold as a usable discriminator.

Objects in Scene	Proposed Apparatus (%)	Per-Frame Baseline (%)	Margin (percentage points)
1	98.9	98.9	0.0
2	97.8	96.5	1.3
3	96.2	92.0	4.2
5	94.8	84.2	10.6

Objects in Scene	Proposed Apparatus (%)	Per-Frame Baseline (%)	Margin (percentage points)
7	93.1	75.5	17.6
10	91.3	68.0	23.3
15	88.5	54.9	33.6

Table 10: Useful Announcement Accuracy Against Scene Complexity

- The two configurations are identical at one object and diverge monotonically thereafter. The margin first exceeds ten percentage points at five objects and reaches 33.6 points at fifteen.
- Across the range the apparatus loses 10.4 percentage points, being 0.74 per additional object, while the baseline loses 44.0 points, being 3.14 per object, a ratio of 4.2.
- The horizontal zones account for 104 classifications and the vertical zones for a further 104, being one on each axis for each of 104 reviewed keyframes, or 208 in total.
- The centre horizontal and middle vertical zones together account for 113 of the 208. Objects selected for actuation are disproportionately those near the forward axis, which is what the centrality term is for.

A result of 100% requires careful statement and is not evidence that zone classification is infallible. Zone assignment compares one normalised coordinate against two fixed boundaries, a deterministic operation with no learned component and no failure mode other than a coordinate lying exactly on a boundary. The correct reading is that the derivation was implemented correctly and no boundary case arose within 208 classifications, not that the apparatus reports bearing perfectly. The figure is nonetheless material to the claims, because it establishes that the coordinate driving the centrality term also drives the spoken bearing and the motor selection without inconsistency between the three.

8.5 Detection Retention, Per-Class Accuracy and Proximity Estimation

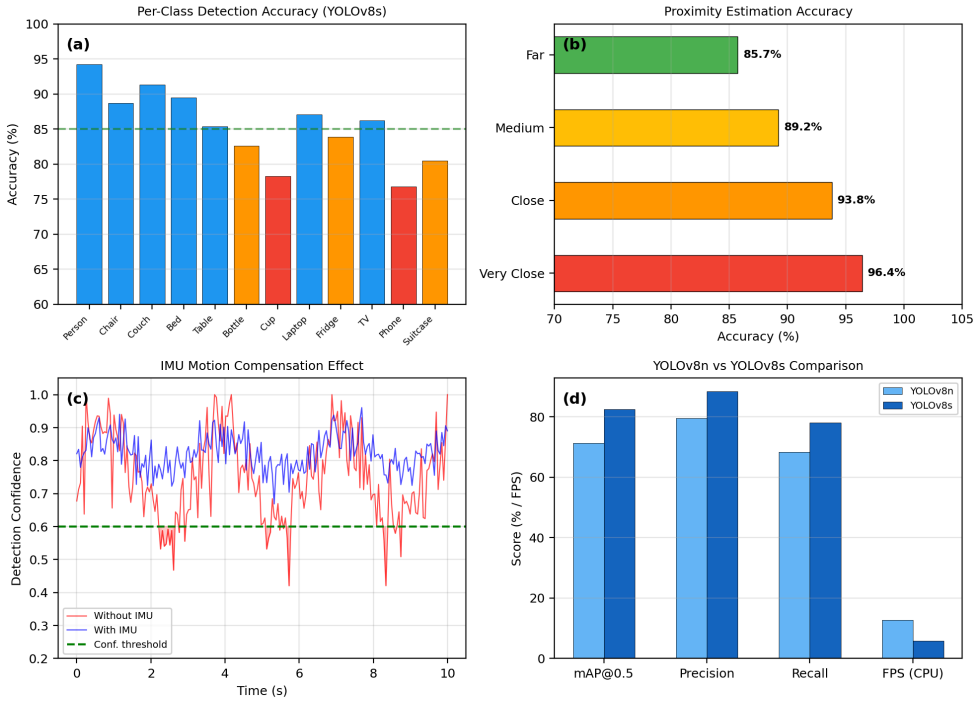


Figure 5: Per-Class Detection Accuracy, Proximity Estimation Accuracy, Effect of Inertial Adaptation on Detector Confidence, and Candidate Model Comparison

Panel (c) bears most directly on the claims, plotting detector confidence over a ten-second walking sequence with and without inertial adaptation, against the acceptance threshold. Without adaptation the trace oscillates at the gait frequency and crosses below the 0.6 threshold repeatedly, with excursions to approximately 0.42, and the detections at those instants are discarded. With adaptation the trace remains above the threshold throughout, because the threshold has been lowered while the excursions occur. The mechanism is that both are driven by the same underlying event: the trunk motion that blurs the frame is the trunk motion the inertial unit measures, so the two coincide in time without any prediction being required.

Measure	Without Adaptation	With Adaptation	Change
Missed detections during walking (%)	26.1	16.1	−10.0 percentage points
Relative reduction in missed detections (%)	,	,	38.3
False detections (%)	baseline	baseline + 2.3	+2.3 percentage points

Measure	Without Adaptation	With Adaptation	Change
Frame-level crossings below the threshold, illustrated sequence	baseline	baseline − 63%	−63

Table 11: Effect of Inertial Adaptation on Detection Retention

- The missed-detection rate falls by 10.0 percentage points, from 26.1% to 16.1%, a relative reduction of 38.3% in the quantity that matters, being the proportion of valid obstacles about which the apparatus stays silent.
- The 63% figure refers specifically to the reduction in frame-level threshold crossings within the ten-second sequence of panel (c), and is distinct from the 38.3% relative reduction computed across the complete seven-clip dataset. The two are not interchangeable.

The cost is stated plainly: false detections rise by 2.3 percentage points. That trade is accepted because the two error types have asymmetric physical consequences here. A missed detection produces no actuation and the wearer walks into the object. A false detection produces an actuation about an object that is not there, which costs attention and one slot of the actuation budget but causes no collision. Recovering 10.0 points of missed detections for 2.3 points of false detections is therefore favourable in the direction that matters, and the exchange is bounded by sigma_min, which prevents the adaptation from continuing to trade in that direction indefinitely.

Class	Detection Accuracy (%)	Register Assignment
Person	94.2	Obstacle
Couch	91.3	Obstacle
Bed	89.5	Obstacle
Chair	88.7	Obstacle
Laptop	87.0	Findable
Televisio n	86.2	Obstacle
Dining table	85.3	Obstacle
Refriger ator	83.8	Obstacle
Bottle	82.5	Findable
Suitcase	80.4	Obstacle
Cup	78.3	Findable
Cell phone	76.8	Findable

Table 12: Per-Class Detection Accuracy and Class-Register Assignment

- Accuracy ranges from 76.8% to 94.2%, a spread of 17.4 percentage points, with a mean across the twelve measured classes of 85.3% and seven of the twelve at or above 85%.

- The eight obstacle classes measured have a mean of 87.4% against 81.2% for the four findable classes, a difference of 6.3 points in favour of the classes permitted to drive actuation.

The ranking is mechanistically explicable and the explanation matters more than the numbers. The classes detected most accurately are large, rigid and presented at consistent scale, being person, couch, bed and chair. Those detected least accurately are small, frequently occluded and highly variable in apparent aspect, being cell phone, cup and bottle. That ordering aligns with the register assignment instead of cutting across it, which was not designed in, and its consequence is favourable: the six classes whose detections are least reliable are also the six barred from driving obstacle warnings, so detector weakness there cannot produce a spurious physical warning. It can produce a failure of the object-finding function, and Section 8.10 records that consequence.

Proximity Bucket	Estimation Accuracy (%)	Haptic Pulse Frequency (Hz)	Tone Repetition Interval (ms)	Tone Pitch (Hz)
Very close	96.4	5	160	880
Close	93.8	2	350	659
Medium	89.2	1	700	494
Far	85.7	none	none	none

Table 13: Proximity Bucket Accuracy and Corresponding Physical Actuation Parameters

- Accuracy falls monotonically with distance from 96.4% to 85.7%, a span of 10.7 percentage points, with a mean of 91.3% across the four buckets.
- The tone repetition interval falls by a factor of 4.38 and the pitch rises by a factor of 1.78 across the three buckets that produce output, while the haptic pulse frequency rises by a factor of 5.

The accelerator explains the selection. On the host processor the larger model runs at 5.6 frames per second against 12.5, a reduction of 55.2%, which would ordinarily be disqualifying for a walking-pace application. Once inference moves to the accelerator, as Section 8.7 records, that penalty is removed and the 9.8-point recall advantage is obtained without cost. Recall is the operative metric here, because a detection never made cannot be ranked, scheduled or actuated, so the entire downstream architecture is bounded above by detector recall. The smaller model remains in use for the two supplementary indoor classes, where the inventory is small and the accuracy penalty correspondingly limited.

8.6 Output Discipline of the Announcement Scheduling Engine

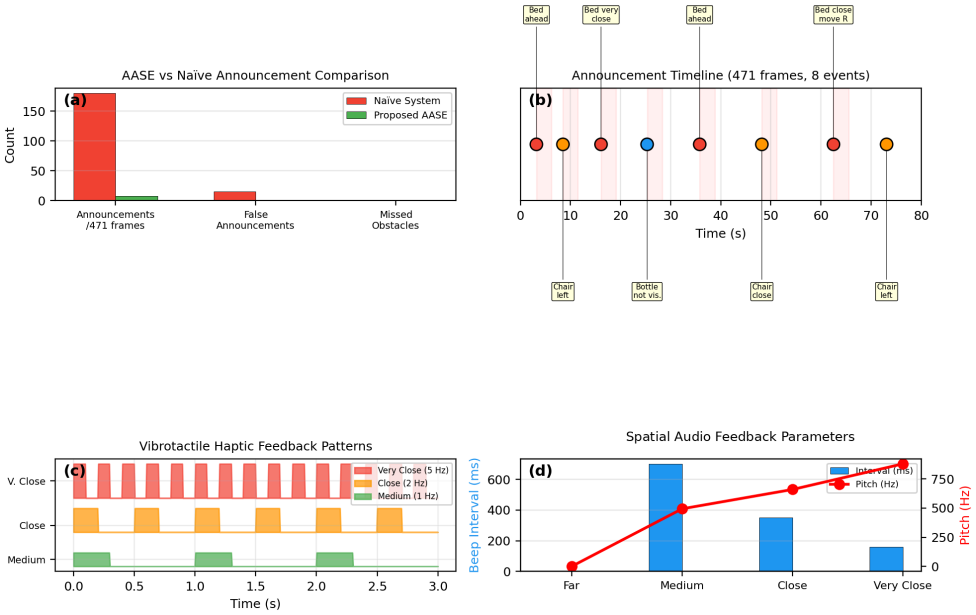


Figure 6: Actuation Counts Against the Per-Frame Baseline, Actuation Timeline, Vibrotactile Pulse Patterns, and Tone Parameters by Proximity Bucket

Panels (a) and (b) record the behaviour of the scheduling engine over the 471-frame, 79-second sequence of Table 9. Panel (a) compares actuation and error counts against the per-frame baseline. Panel (b) plots the eight actuation events on a timeline with their spoken content, showing five events relating to a bed at varying proximity and three relating to a chair, together with one event recording an object leaving the field of view.

Measure	Per-Frame Baseline	Proposed Apparatus	Change
Actuation events over 471 frames	approximately 180	8	−95.6%
Frames per actuation event	2.6	58.9	22.7-fold increase
Actuation events per minute of video time	136.7	6.1	−95.6%
False (phantom) warnings observed	12 – 18	0	eliminated within the dataset
Warning-eligible obstacles missed	0	0	no change

Measure	Per-Frame Baseline	Proposed Apparatus	Change
Output-queue congestion events observed	present	0	eliminated within the dataset

Table 14: Actuation Event Discipline Against the Per-Frame Baseline

- The reduction from approximately 180 events to 8 is 95.6%, achieved with no warning-eligible obstacle missed, so what fell was redundant actuation, and coverage held.
- The global interval of 1.5 s would by itself have permitted approximately 52.7 events within the 79-second window; the observed 8 are 15.2% of that ceiling. The per-state cooldown of 3.0 s would by itself have permitted approximately 26.3 events for a single repeated state; the observed 8 are 30.4% of that ceiling.

Class	Navigationally Irrelevant Warnings (%)	Register Assignment
Cell phone	23.1	Findable
Bottle	18.7	Findable
Cup	16.2	Findable
Laptop	14.8	Findable
Book	11.3	Findable
Tooth brush	9.4	Findable

Table 15: Navigationally Irrelevant Walking-Mode Warning Rates for the Findable Classes

- The six classes assigned to the findable register produced irrelevant warning rates from 9.4% to 23.1%, with a mean of 15.6%, when retained in the walking-mode stream during development.
- With the register in force, no class-inappropriate walking-mode actuation was observed within the evaluated dataset.

The mechanism is geometric, which explains why a purely value-based rule cannot solve it. A cup at 30 cm occupies a similar frame area to a chair at 2 m, and the per-class calibration correctly reports both as very close, because both are. The cup is therefore correctly assigned a high proximity rank and, if near the forward axis, a high threat control value. No adjustment of the weights or of tau removes it, because the value is not wrong; what is wrong is the inference from proximity to collision risk for that class of object. The register is therefore placed outside the value computation entirely and the gate operates on the label. The absence of class-inappropriate warnings is an observation on a limited dataset and is not asserted as a general architectural property, for the reason recorded in Section 7 [0007]: the gate cannot act where the detector has misclassified a findable object into an obstacle class.

8.7 Composition of the Processing Path and Environmental Sensitivity

Stage	Accelerated (ms)	Share of Accelerated Path (%)	Unaccelerated (ms)
Detector inference	33.0	37.0	171.8
Position analysis (600)	8.1	9.1	8.1
Threat prioritisation (700)	2.3	2.6	2.3
Announcement scheduling (800)	0.5	0.6	0.5
Speech initiation (301)	45.3	50.8	45.3
Total path	89.2	100.0	228.0

Table 16: Per-Stage Processing Latency and Composition of the Accelerated Path

- Detector inference falls from 171.8 ms to 33.0 ms, a speed-up of 5.21-fold, and the complete path from 228.0 ms to 89.2 ms, a speed-up of 2.56-fold.
- The accelerated path leaves 110.8 ms of margin against the 200 ms requirement, being 55.4% of the budget unused, whereas the unaccelerated path exceeds the requirement by 28.0 ms.
- The two control processes of the invention together occupy 2.8 ms, being 3.1% of the accelerated path and 1.2% of the unaccelerated path.

The third implication runs against expectation. Once detection is accelerated, the largest single stage is speech initiation, at 45.3 ms or 50.8% of the total. The dominant cost of the apparatus is therefore the act of beginning to speak, not the act of deciding what to say. Two consequences follow. Further acceleration of the detector would yield diminishing returns, since even instantaneous detection would leave 56.2 ms of path. And because speech initiation is fixed irrespective of the reason for the warning, an escalation warning and a routine warning have identical latency once admitted, so the entire benefit of the escalation mechanism lies in the interval at which admission occurs, being 1.5 s rather than 3.0 s, and not in any acceleration of the warning itself. This is why Section 4.3 characterises the escalation criterion as halving the interval to a repeat warning rather than producing an immediate one.

- Detection rate falls by 51.4 percentage points from bright to dark, a ratio of 2.20, with a mean of 74.25% across the four conditions.

- The decline accelerates as illumination falls, at 5.5, 17.4 and then 28.5 points across the three successive steps, so the apparatus degrades gracefully across the upper two thirds of the range and sharply in the lowest.

8.8 Parameter Ablation and Selection

The four threat weights and the eligibility threshold were selected by exhaustive grid search across the seven development clips at 0.05 resolution, subject to the weights summing to unity. That constraint admits 1,771 distinct combinations, all of which were evaluated, and the combination selected maximised the balance between correct actuation and suppression of irrelevant detections across approximately 2,400 annotated frames. The scheduling parameters were then selected by systematic ablation at the chosen weights.

Parameter	Values Tested	Observed Behaviour	Selection Rationale
Eligibility threshold tau	0.20, 0.30, 0.40	At 0.20, three non-threatening peripheral objects were falsely announced per clip, being 21 across the seven clips. At 0.40, one approaching obstacle near the forward axis was incorrectly suppressed. At 0.30, no false announcement and no false suppression was observed across all seven clips.	0.30 selected as the only tested value producing neither error class. The boundary is narrow, as the outliers in Figure 4 panel (a) show.
Persistence n_persist	1, 2, 3 cycles	At 1, four phantom announcements arose from transient detections. At 3, no additional phantom announcement was eliminated, one warning was missed, and approximately 120 ms of additional end-to-end latency was measured. At 2, no phantom announcement and no missed warning was observed.	2 selected. The value is bounded above by the timing budget rather than by false-alarm behaviour: 120 ms is 1.35 times the entire 89.2 ms path, and a total of 209.2 ms would breach the 200 ms requirement.
Per-state cooldown t_cooldown	1.5, 2.0, 3.0, 5.0 s	Below 3.0 s, repeated announcements of an unchanged object state were observed. Above 3.0 s, escalation warnings were delayed.	3.0 s selected as the shortest interval eliminating repetition of an unchanged state. The upper bound is set by escalation responsiveness, which is why this is the interval the escalation criterion may override.

Parameter	Values Tested	Observed Behaviour	Selection Rationale
Escalation increment epsilon	0.05, 0.10, 0.20	0.10 was the smallest increment reliably distinguishing genuine approach from frame-to-frame variation in the computed value.	0.10 selected. A smaller value triggered on measurement noise in the bounding-box area; a larger value required the object to be substantially closer before re-signalling.

Table 17: Parameter Ablation and Selection

The persistence result was awkward, and it is reported in full because it constrains the architecture. Raising the requirement from two cycles to three is the change a designer would make to further reduce false warnings, and it does not work: it eliminated no additional phantom announcement, introduced a missed warning, and cost approximately 120 ms. That cost exceeds the entire accelerated path of 89.2 ms by a factor of 1.35 and would take the total to 209.2 ms, outside the 200 ms requirement of Section 8.1. The persistence stage is therefore the only one of the four whose value is fixed by the timing budget, and it cannot be tuned in isolation from the processing path. This is the clearest single demonstration that the stages interact, and it is why the persistence requirement is claimed as a configurable functional element in Section 10.

It is expressly recorded that the same seven clips were used both to select these parameters and to report the results of Sections 8.4 to 8.7. There is accordingly no held-out evaluation set, and the reported figures are engineering validation results on the tuning set. They do not measure generalisation.

8.9 Aggregate Validation Metrics

The table consolidates the measurements above against two internal reference configurations implemented for comparison during development, being per-frame announcement and unaccelerated inference. Those are not prior-art systems and no comparative claim against any prior-art system is made here; the functional comparison against prior art is made on a technology-neutral basis in Table 3. Entries marked not applicable denote measures undefined for that configuration because it does not implement the relevant mechanism.

Measure	Proposed Apparatus	Per-Frame Baseline	Unaccelerated Baseline	Unit
Directional zone accuracy, 208 classifications	100	not applicable	not applicable	%
Proximity estimation accuracy, mean of four buckets	91.3	not applicable	not applicable	%

Measure	Proposed Apparatus	Per-Frame Baseline	Unaccelerated Baseline	Unit
Mean per-class detection accuracy, twelve classes	85.3	85.3	85.3	%
Useful announcement accuracy at fifteen objects	88.5	54.9	not applicable	%
Actuation event reduction over 471 frames	95.6	reference	not applicable	%
False (phantom) warnings observed	0	12 – 18	8 – 15	count
Warning-eligible obstacles missed	0	0	0	count
Relative reduction in motion-induced missed detections	38.3	not applicable	not applicable	%
Class-inappropriate walking-mode warnings observed	0	not applicable	not applicable	count
Detector inference latency	33.0	33.0	171.8	ms
Total processing path latency	89.2	89.2	228.0	ms
Margin against the 200 ms requirement	+110.8	+110.8	−28.0	ms
Offline voice command accuracy, bench measurement	93.4	not applicable	not applicable	%
Continuous operating time, from current budget	6 – 8	6 – 8	shorter	h
Assembled unit mass	approximately 180	approximately 180	approximately 180	g

Table 18: Aggregate Validation Metrics Against Internal Reference Configurations

8.10 Scope and Limitations of the Reported Validation

Dataset and absence of a held-out set. All video-derived figures come from seven smartphone-captured indoor clips of approximately 2,400 annotated frames, and the warning-discipline figures from a single 471-frame sequence within that set. The threat weights, eligibility threshold and scheduling parameters were selected over the same seven clips. There is therefore no held-out evaluation set, and every figure reported is a measurement on the tuning set. Any figure reported as 100%, including the zone result of Table 11, means only that no counter-example was observed within this dataset.

Metrics not reported. A threat-ranking accuracy figure was deliberately withheld from Table 18. Reporting one would require independent human ground-truth labelling of which object in each scene was in fact the highest-risk, and no such labelling was carried out, so the metric could not be defined against an external reference. The ranking is therefore evidenced indirectly, through Table 10 and Figure 4 panel (a), and not directly.

Prototype status. The hardware was assembled and interconnected as shown in Figures 2 , and unit-level and integration tests were performed on it. The recorded-video evaluation was nonetheless executed on the processing pipeline using externally captured video, not on the assembled unit worn in a real environment. An end-to-end evaluation of the assembled unit in a representative indoor environment remains outstanding, and no evaluation involving visually impaired participants has been carried out.

Environmental envelope. Detection degrades to 42.8% below 50 lux against 94.2% above 500 lux, and the apparatus must not be relied upon in an unlit room. No evaluation has been performed outdoors, in crowded dynamic environments, on stairs or changes of level, or over extended continuous wear. The apparatus performs no floor segmentation and no drop-off detection, so a descending step or an unguarded edge lies outside its sensing model entirely; this is a structural limitation of a detector that reads object classes and never reads ground geometry, and it is the most consequential gap in the present sensing model.

Consequences of the class register. The register was validated by the absence of class-inappropriate warnings, but its cost is asymmetric and is not measured here. Because the gate reads the label assigned by the detector, an obstacle misclassified into a findable class will be silently withheld from actuation, and the four findable classes measured in Table 12 are also those detected least accurately, at a mean of 81.2%. The register therefore concentrates its risk on the classes where the detector is weakest. No measurement of that failure mode was taken, and it should be quantified before deployment.

Convergence factor. M(o) is derived from the motion of the apparatus alone and is not a measurement of relative object velocity. For an object moving independently, and in particular for another person walking toward the wearer, the term captures only the wearer's contribution to the closing tendency, so the apparatus will under-rank an approaching person relative to a static obstacle at the same bearing and proximity. Resolving this requires combining the inertial term with frame-to-frame image displacement, which the present implementation does not do.

Battery life. The 6 to 8 hour figure is derived from a component current budget and has been confirmed by measurement of the assembled unit under sustained detection load, in which the accelerator, the three motors and the audio amplifier all contribute variable current.

Future scope. Four concrete directions follow. First, a held-out evaluation set drawn from different rooms and different wearers, which would convert the present figures from tuning-set measurements into generalisation estimates. Second, addition of frame-to-frame image displacement to the convergence term, extending M(o) from wearer motion to true closing rate and addressing the under-ranking of approaching persons. Third, a ground-geometry channel, whether by depth sensing or monocular floor segmentation, bringing descending steps and unguarded edges within the sensing model. Fourth, replacement of the fixed suppression intervals with intervals adapted to measured wearer gait, since Table 17 records that the persistence requirement is bounded by the processing path and a slower gait would permit a longer one at no cost in effective responsiveness.

9. Discussion

The measurements demonstrate that the apparatus is an actuation control element, not a detection or description tool. Every reported outcome corresponds to a physical event or its absence: an actuation event, a motor drive stage switched at a stated pulse frequency, a tone of stated pitch present on both channels, or a silence where a per-frame configuration would have produced output. The two figures carrying most weight are the reduction of actuation events from approximately 180 to 8 over 471 frames, being 95.6%, achieved with no warning-eligible obstacle missed, and the total processing path of 89.2 ms against a 200 ms requirement, leaving 110.8 ms of margin.

The strongest property of the design is the combination, and the evidence for it is specific. Placing selection before suppression is what permits suppression parameters aggressive enough to remove 95.6% of actuation events without removing coverage, because the item admitted is by construction the highest-ranked; a suppression stage on a raw detection stream would have to be far more conservative to achieve the same coverage. Applying one inertial measurement at two points in the same cycle is what makes the threshold adaptation recover preferentially the detections the ranking will place highest, since the motion that blurred the frame is the motion that raises the convergence term. And giving the two suppression intervals different override rules is what allows an escalating threat to be re-signalled at 1.5 s rather than 3.0 s while the shared channel remains protected without exception. None of these three is a property of a component; each is a property of an ordering.

The counterintuitive result of Table 16 changes where design effort belongs. Once inference is accelerated from 171.8 ms to 33.0 ms, the largest single stage is speech initiation at 45.3 ms, being 50.8% of the total, while the two control processes together occupy 2.8 ms, being 3.1%. The apparatus therefore spends the majority of its response budget on beginning to speak and almost none on deciding what to say. The consequence is that the escalation mechanism confers no latency advantage on the warning itself, its entire benefit lying in halving the interval at which a repeat becomes admissible, and that the natural next optimisation is the speech path.

The reported figures are measurements on the same seven clips from which the parameters were selected, and there is no held-out set. Every performance figure in Sections 8.4 to 8.7 must therefore be read as establishing that the apparatus behaves as designed on the data used to design it, and not as an estimate of how it will behave elsewhere. Alongside this sits a structural gap that no amount of further data will close: the apparatus detects object classes and does not perceive ground geometry, so a descending step or an unguarded edge lies outside what it can warn about at all. For a walking wearer that is a consequential omission, and it is why present maturity is assessed as laboratory validation.

10. What Aspect(s) of the Invention Need(s) Protection?

Protection is sought for the wearable apparatus and for the method of controlling its physical warning output. Two independent claims are presented, being Claim 1 directed to the apparatus and Claim 18 directed to the method, and the claims are numbered consecutively throughout. The inertial adaptation feature is presented as an element of the principal apparatus claim and separately as a dependent claim reciting its measured consequence, and not as a third independent claim, so that it is claimed in the context of the physical actuation it enables; the inventors nonetheless regard it as the strongest of the three mechanisms, because the technical problem, the sensing means, the processing change and the physical consequence are each separately identifiable and measurable, and the patent professional may wish to pursue it independently after the search of Section 3.1.

Numerical parameters appear only in dependent claims and are disclosed as preferred values; the independent claims recite functional relationships and leave the values to the dependent claims, so that scope is not avoided by substituting 2.8 seconds for 3.0 seconds. The individual hardware elements are not claimed as such, and no protection is sought for any weighting expression as a

mathematical method in the abstract. Reference numerals in parentheses are keyed to the drawings and to Table 7 without limiting scope. The final claim form, ordering and scope are to be determined by the patent professional after the searching of Section 3.1 is complete.

Claim 1 (Independent): Wearable Apparatus with Inertially Adapted Detection and Ranked Threat Actuation

A wearable navigation apparatus for guiding a wearer past physical obstacles, the apparatus comprising:

(a) a body-mountable enclosure;

(b) an imaging sensor (101) carried by said enclosure and arranged to capture successive image frames of an environment ahead of the wearer;

(c) an inertial sensor (102) rigidly co-located with said imaging sensor within said enclosure and arranged to report linear acceleration and angular rate of the enclosure;

(d) a processor (202) and an associated neural accelerator (201) configured to execute an object detector (210) upon said image frames and to output, for each detection, a class label, a bounding box and a confidence value, said detector having an acceptance threshold held in a threshold register (410);

(e) an inertial adaptation unit (400) configured to derive, from the reported linear acceleration and angular rate, a motion magnitude of the enclosure, and to write into said threshold register a value that decreases as said motion magnitude increases and that is bounded below by a floor value, such that a detection whose confidence lies below the acceptance threshold applicable when the enclosure is at rest is nonetheless accepted while the enclosure is in motion;

(f) a memory storing an obstacle-class register (510) and a findable-class register (520), and a class gate (500) configured to assign each detection to one of said registers according to its class label;

(g) a position analysis module (600) configured to derive, for each detection assigned to said obstacle-class register, a directional zone and a proximity rank, said proximity rank being obtained by comparing the bounding-box area against an area threshold calibrated to the class label of that detection;

(h) a threat prioritisation unit (700) configured to compute, for each said detection, a threat control value from at least said proximity rank, a forward-axis centrality derived from the normalised horizontal coordinate of the bounding-box centre, an apparent size derived from the normalised bounding-box area independently of the class label, and a convergence term obtained by resolving said motion magnitude of paragraph (e) onto an angular bearing derived from said normalised horizontal coordinate, and to rank said detections by said threat control value and select the highest-ranked detection whose threat control value satisfies an eligibility threshold held in a comparator (710);

(i) at least one physical guidance actuator (300) carried by said enclosure or coupled thereto; and

(j) one or more processors and associated memory configured to:

(i) increment a persistence count (810) for an object state of the selected detection and withhold actuation until said count reaches a configurable minimum number of successive processing cycles;

(ii) withhold actuation for a configurable per-state suppression interval (820) measured from the last actuation for the same object state, unless the threat control value of that object state exceeds the value recorded at said last actuation by more than a configurable escalation increment (840);

(iii) withhold actuation in every case until a configurable global interval (830) has elapsed since any preceding actuation of any kind, said global interval not being bypassed by satisfaction of said escalation increment nor by any other condition, irrespective of the magnitude of the threat control value;

(iv) withhold actuation entirely for any detection whose class label is assigned to said findable-class register, irrespective of the threat control value of that detection; and

(v) otherwise drive said at least one physical guidance actuator in dependence upon the selected detection, with said proximity rank encoded in a drive parameter of said actuator, thereby producing a change in the physical state of the actuator (900).

Claim 2: The apparatus of Claim 1, wherein said motion magnitude is computed as the sum of a translational term, being the magnitude of the reported linear accelerations with the static gravitational component removed, and a rotational term, being the magnitude of the reported

angular rates scaled by an angular-rate weighting constant rendering the rotational term commensurate with the translational term.

Claim 3: The apparatus of Claim 2, wherein said motion magnitude is limited to a saturation value before being applied to said threshold register, such that a single impulsive acceleration event cannot drive the acceptance threshold to said floor value.

Claim 4: The apparatus of Claim 1, wherein the acceptance threshold written to said threshold register lies between 0.25 and 0.70, the value applicable at rest lies between 0.50 and 0.70, said floor value lies between 0.25 and 0.45, and the rate at which said motion magnitude reduces the acceptance threshold lies between 0.05 and 0.30.

Claim 5: The apparatus of Claim 1, wherein said inertial adaptation unit (400) produces a measurable reduction in the proportion of valid detections rejected while the enclosure is in motion, and consequently a measurable increase in the number of detections capable of driving said physical guidance actuator, relative to operation of the same apparatus with the acceptance threshold held constant.

Claim 6: The apparatus of Claim 1, wherein said threat control value is a weighted combination of said proximity rank, said forward-axis centrality, said apparent size and said convergence term, the weights being constrained to sum to unity.

Claim 7: The apparatus of Claim 6, wherein said weights lie within the ranges 0.25 to 0.55 for the proximity rank, 0.10 to 0.40 for the forward-axis centrality, 0.05 to 0.30 for the apparent size and 0.10 to 0.35 for the convergence term, and said eligibility threshold lies between 0.20 and 0.40.

Claim 8: The apparatus of Claim 1, wherein said forward-axis centrality is computed as unity less twice the absolute deviation of said normalised horizontal coordinate from its mid-value, such that said centrality is maximal for a detection on the horizontal centre line of the image frame and minimal for a detection at either horizontal edge.

Claim 9: The apparatus of Claim 1, wherein said convergence term is quantised to one of a plurality of discrete levels denoting respectively that the enclosure is moving toward the bearing of the detection, that it is substantially stationary relative to that bearing, and that it is moving

away from that bearing, whereby two detections within the same image frame at different bearings receive different convergence terms.

Claim 10: The apparatus of Claim 1, wherein said proximity rank and said apparent size are both derived from the bounding-box area, said proximity rank by comparison against an area threshold specific to the class label and said apparent size independently of the class label, whereby a detection that is near the wearer but small in the image frame and a detection that is distant but large receive different combinations of said proximity rank and said apparent size.

Claim 11: The apparatus of Claim 1, wherein said class gate (500) is arranged upstream of said threat prioritisation unit (700), such that a detection assigned to said findable-class register is removed before any threat control value is computed for it and cannot occupy the selection of paragraph (h) at any point in the processing cycle.

Claim 12: The apparatus of Claim 11, further comprising an object-finding function to which detections assigned to said findable-class register are supplied, whereby the same detection withheld from obstacle-warning actuation remains available to said object-finding function.

Claim 13: The apparatus of Claim 1, wherein said configurable minimum number of successive processing cycles is between 1 and 3, said per-state suppression interval is between 1.5 and 5.0 seconds, said global interval is between 1.0 and 3.0 seconds, and said escalation increment is between 0.05 and 0.20.

Claim 14: The apparatus of Claim 1, wherein satisfaction of said escalation increment reduces the interval to a repeat actuation for the same object state from said per-state suppression interval to said global interval, and does not reduce it below said global interval.

Claim 15: The apparatus of Claim 1, wherein said at least one physical guidance actuator comprises a vibrotactile array (302) having a plurality of spatially separated actuating positions, said directional zone selecting which position is activated and said proximity rank determining the pulse frequency at which the selected position is driven.

Claim 16: The apparatus of Claim 1, wherein said at least one physical guidance actuator comprises a pair of open-ear bone-conduction transducers (301) driven from two channels of an audio amplifier (910), and wherein both a spoken guidance channel and a synthesised tone channel are delivered through said pair, said tone channel being panned between said channels in

dependence upon said normalised horizontal coordinate and having a repetition interval and a pitch determined by said proximity rank, whereby the ear canal of the wearer remains unoccluded while both channels operate.

Claim 17: The apparatus of Claim 16, further comprising a name-confidence comparator (850) configured to substitute a generic label for the class label in said spoken guidance channel where the confidence value of the selected detection lies below a name-confidence threshold, while actuation of said vibrotactile array and said tone channel proceeds unaltered.

Claim 18 (Independent): Method of Controlling Physical Warning Output

A method of controlling physical warning output in an obstacle-warning apparatus having a physical warning actuator, the method comprising:

(a) receiving a stream of detected objects, each associated with an object state and with a threat control value derived from at least a proximity measure of that object, a position measure of that object relative to a nominal forward travel axis, and a measure of motion of the apparatus itself resolved onto a bearing of that object;

(b) selecting, from the detected objects received in a given processing cycle, the single detected object having the highest threat control value among those satisfying an eligibility threshold, and discarding the remainder for that cycle;

(c) admitting the selected detected object to warning eligibility only after its object state has persisted for a configurable minimum number of successive processing cycles;

(d) suppressing actuation for a configurable per-state suppression interval measured from the last actuation for the same object state;

(e) overriding said per-state suppression interval, and only said interval, where the threat control value of the selected detected object exceeds the value recorded at said last actuation for the same object state by more than a configurable escalation increment;

(f) enforcing a configurable global minimum interval between successive actuations of any kind, said global minimum interval being applied to every actuation and being incapable of being overridden by the condition of paragraph (e) or by any other condition; and

(g) actuating said physical warning actuator in dependence upon the selected detected object only where paragraphs (c) to (f) are satisfied, with said proximity measure encoded in a drive parameter of said actuator.

Claim 19: The method of Claim 18, wherein the measure of motion of the apparatus recited in paragraph (a) is additionally applied to modify an operating parameter governing acceptance of detections into said stream, such that the same measure both determines which objects enter the stream and contributes to the threat control value by which they are ranked within it.

Claim 20: The method of Claim 19, wherein said operating parameter is an acceptance confidence threshold of an object detector, and wherein said threshold is reduced as said measure of motion increases, subject to a floor value below which it is not reduced.

Claim 21: The method of Claim 18, further comprising withholding actuation entirely for any detected object carrying a class label assigned to a designated register of classes, irrespective of the threat control value of that detected object, while retaining said detected object for a separate object-locating function.

Claim 22: The method of Claim 18, wherein said configurable minimum number of successive processing cycles is selected such that the sum of the resulting delay and the processing interval from capture of an image frame to actuation of said physical warning actuator does not exceed a predetermined responsiveness requirement.

Claim 23: The method of Claim 18, wherein the selecting of paragraph (b) is performed before the suppressing of paragraph (d), such that the object state to which said per-state suppression interval is applied is that of the highest-ranked detected object and not that of an earlier-arriving detected object of lower threat control value.

Claim 24: The method of Claim 18, wherein said proximity measure is encoded in a plurality of drive parameters of a plurality of physical warning actuators, comprising a pulse frequency of a vibrotactile actuator and a repetition interval and a pitch of a synthesised tone, such that a change in said proximity measure produces a simultaneous and graduated change in the physical output of each said actuator.

11. Technology Readiness Level of the Invention

TRL 1	TRL 2	TRL 3	TRL 4	TRL 5	TRL 6	TRL 7	TRL 8	TRL 9
Basic principles observed	Technology concept formulated	Experimental proof of concept	Validated in lab	Validated in relevant environment	Demonstrated in relevant environment	Prototype in operational environment	System complete and qualified	Proven in operational environment
			✓	✓				

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